STUDY GUIDE

Study Plans and Progress Checklist

7-, 14-, 21- and 30-day plans on the minimum-viable and complete paths

Pick the plan that matches your time, tick lessons as you go, and use the mastery check column honestly.

Generated 10 October 2026 from the same lesson data that powers the website.
Every Java program in this guide was compiled with --release 17 and run against its expected output.

Study plans

Plans pack lessons in curriculum order (so prerequisites always come first), revise yesterday's and the previous days' material, and finish with a weak-topic review and a mixed assessment. The website rebuilds the plan from your actual results.

Minimum-viable path: 55 lessons (about 36 hours of reading plus practice) covering every essential syntax and core idea. **Complete path:** all 83 lessons (about 56 hours of reading).

7-day plan — minimum-viable path

Day	What to do	Minutes
1	Learn: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Practise: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Learn: 0.3 Installing Java on macOS and Choosing an Editor · Practise: 0.3 Installing Java on macOS and Choosing an Editor · Learn: 0.4 Your First Program: class, main, statements, comments · Practise: 0.4 Your First Program: class, main, statements, comments · Learn: 0.5 Compile-time, Runtime and Logic Errors · Practise: 0.5 Compile-time, Runtime and Logic Errors · Learn: 0.6 What Is an Algorithm? · Practise: 0.6 What Is an Algorithm? · Learn: 1.1 Variables, Constants and Primitive Types · Practise: 1.1 Variables, Constants and Primitive Types · Learn: 1.2 Type Conversion, Casting and Overflow · Practise: 1.2 Type Conversion, Casting and Overflow · Learn: 1.3 Arithmetic, Assignment and Increment Operators · Practise: 1.3 Arithmetic, Assignment and Increment Operators · Learn: 1.4 Relational, Logical and Ternary Operators; Precedence · Practise: 1.4 Relational, Logical and Ternary Operators; Precedence · Learn: 1.5 Bitwise Operators and Shifts · Practise: 1.5 Bitwise Operators and Shifts · Learn: 1.6 Output: print, println, printf and String Concatenation · Practise: 1.6 Output: print, println, printf and String Concatenation	574
2	Revise yesterday: How Java Works: Source, Bytecode, JVM, JRE, JDK, Installing Java on macOS and Choosing an Editor, Your First Program: class, main, statements, comments… · Learn: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Practise: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Learn: 1.8 Decision Making: if, else, switch · Practise: 1.8 Decision Making: if, else, switch · Learn: 1.9 Loops: for, while, do-while, break and continue · Practise: 1.9 Loops: for, while, do-while, break and continue · Learn: 1.10 Methods: Parameters, Return Values, Scope and Overloading · Practise: 1.10 Methods: Parameters, Return Values, Scope and Overloading · Learn: 1.11 Tracing and Debugging Programs · Practise: 1.11 Tracing and Debugging Programs · Learn: 2.1 One-Dimensional Arrays · Practise: 2.1 One-Dimensional Arrays · Learn: 2.2 Two-Dimensional Arrays (Matrices) · Practise: 2.2 Two-Dimensional Arrays (Matrices) · Learn: 2.3 Strings, StringBuilder and Characters · Practise: 2.3 Strings, StringBuilder and Characters · Learn: 2.4 Number Systems, Binary and Two's Complement · Practise: 2.4 Number Systems, Binary and Two's Complement	571
3	Revise yesterday: Input: Scanner, BufferedReader and Command-Line Arguments, Decision Making: if, else, switch, Loops: for, while, do-while, break and continue… · Learn: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic · Practise: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic · Learn: 2.6 Recursion: Base Case, Recursive Case and the Call Stack · Practise: 2.6 Recursion: Base Case, Recursive Case and the Call Stack · Learn: 2.7 References, Pass-by-Value and Memory · Practise: 2.7 References, Pass-by-Value and Memory · Learn: 2.8 Boundary Conditions, Validation and Common Mistakes · Practise: 2.8 Boundary Conditions, Validation and Common Mistakes · Learn: 3.8 Generics and Lists: ArrayList and LinkedList · Practise: 3.8 Generics and Lists: ArrayList and LinkedList · Learn: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Practise: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Learn: 3.10 Queues, Deques, Stacks and PriorityQueue · Practise: 3.10 Queues, Deques, Stacks and PriorityQueue · Learn: 3.11 Sorting with Comparable and Comparator · Practise: 3.11 Sorting with Comparable and Comparator · Learn: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Practise: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables	581

Day	What to do	Minutes
4	Revise yesterday: Divisibility, GCD/LCM, Primes and Modular Arithmetic, Recursion: Base Case, Recursive Case and the Call Stack, References, Pass-by-Value and Memory… · Spaced review (3 days ago): How Java Works: Source, Bytecode, JVM, JRE, JDK, Installing Java on macOS and Choosing an Editor · Learn: 4.3 Searching: Linear and Binary Search · Practise: 4.3 Searching: Linear and Binary Search · Learn: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability · Practise: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability · Learn: 4.5 Merge Sort and Divide and Conquer · Practise: 4.5 Merge Sort and Divide and Conquer · Learn: 4.7 Time Complexity and Big-O Notation · Practise: 4.7 Time Complexity and Big-O Notation · Learn: 4.8 Algorithm Design Paradigms · Practise: 4.8 Algorithm Design Paradigms · Learn: 4.9 Analysing Loops: Counting Iterations · Practise: 4.9 Analysing Loops: Counting Iterations · Learn: 4.11 Space Complexity and the Call Stack · Practise: 4.11 Space Complexity and the Call Stack · Learn: 5.1 Euclid's Algorithm (GCD) · Practise: 5.1 Euclid's Algorithm (GCD) · Learn: 5.2 Euler's Phi (Totient) Function · Practise: 5.2 Euler's Phi (Totient) Function	624
5	Revise yesterday: Searching: Linear and Binary Search, Elementary Sorting: Bubble, Insertion, Selection; Stability, Merge Sort and Divide and Conquer… · Spaced review (3 days ago): Input: Scanner, BufferedReader and Command-Line Arguments, Decision Making: if, else, switch · Learn: 5.7 Simple Sieve of Eratosthenes · Practise: 5.7 Simple Sieve of Eratosthenes · Learn: 5.9 Binary Palindrome · Practise: 5.9 Binary Palindrome · Learn: 5.12 Josephus Problem · Practise: 5.12 Josephus Problem · Learn: 5.13 Block Swap Algorithm (Array Rotation) · Practise: 5.13 Block Swap Algorithm (Array Rotation) · Learn: 5.14 Leaders in an Array · Practise: 5.14 Leaders in an Array · Learn: 5.15 Majority Element (Boyer–Moore Voting) · Practise: 5.15 Majority Element (Boyer–Moore Voting) · Learn: 5.16 Maximum Product Subarray · Practise: 5.16 Maximum Product Subarray · Learn: 5.17 Maximum Equilibrium Sum · Practise: 5.17 Maximum Equilibrium Sum · Learn: 5.18 Maximum Sum of an Hourglass in a Matrix · Practise: 5.18 Maximum Sum of an Hourglass in a Matrix	604
6	Revise yesterday: Simple Sieve of Eratosthenes, Binary Palindrome, Josephus Problem… · Spaced review (3 days ago): Divisibility, GCD/LCM, Primes and Modular Arithmetic, Recursion: Base Case, Recursive Case and the Call Stack · Learn: 5.19 Toggle the Switch (100 Doors) · Practise: 5.19 Toggle the Switch (100 Doors) · Learn: 5.22 Move Hyphen to the Beginning · Practise: 5.22 Move Hyphen to the Beginning · Learn: 5.24 Lexicographically First Palindromic Rearrangement · Practise: 5.24 Lexicographically First Palindromic Rearrangement · Learn: 5.27 Maneuvering (Counting Grid Paths) · Practise: 5.27 Maneuvering (Counting Grid Paths) · Learn: 5.29 N-Queens (Backtracking) · Practise: 5.29 N-Queens (Backtracking) · Learn: 5.30 Maze Solving (Rat in a Maze) · Practise: 5.30 Maze Solving (Rat in a Maze) · Learn: 5.33 Selection Sort · Practise: 5.33 Selection Sort · Learn: 5.34 Quick Sort · Practise: 5.34 Quick Sort	570
7	Weak-topic review (from your quiz and practice results) · Final mixed assessment (archive topics)	75

7-day plan — complete path

Day	What to do	Minutes
1	Learn: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Practise: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Learn: 0.2 The Java Landscape: History, Features, Editions, JVM Memory · Practise: 0.2 The Java Landscape: History, Features, Editions, JVM Memory · Learn: 0.3 Installing Java on macOS and Choosing an Editor · Practise: 0.3 Installing Java on macOS and Choosing an Editor · Learn: 0.4 Your First Program: class, main, statements, comments · Practise: 0.4 Your First Program: class, main, statements, comments · Learn: 0.5 Compile-time, Runtime and Logic Errors · Practise: 0.5 Compile-time, Runtime and Logic Errors · Learn: 0.6 What Is an Algorithm? · Practise: 0.6 What Is an Algorithm? · Learn: 1.1 Variables, Constants and Primitive Types · Practise: 1.1 Variables, Constants and Primitive Types · Learn: 1.2 Type Conversion, Casting and Overflow · Practise: 1.2 Type Conversion, Casting and Overflow · Learn: 1.3 Arithmetic, Assignment and Increment Operators · Practise: 1.3 Arithmetic, Assignment and Increment Operators · Learn: 1.4 Relational, Logical and Ternary Operators; Precedence · Practise: 1.4 Relational, Logical and Ternary Operators; Precedence · Learn: 1.5 Bitwise Operators and Shifts · Practise: 1.5 Bitwise Operators and Shifts · Learn: 1.6 Output: print, println, printf and String Concatenation · Practise: 1.6 Output: print, println, printf and String Concatenation · Learn: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Practise: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Learn: 1.8 Decision Making: if, else, switch · Practise: 1.8 Decision Making: if, else, switch · Learn: 1.9 Loops: for, while, do-while, break and continue · Practise: 1.9 Loops: for, while, do-while, break and continue · Learn: 1.10 Methods: Parameters, Return Values, Scope and Overloading · Practise: 1.10 Methods: Parameters, Return Values, Scope and Overloading · Learn: 1.11 Tracing and Debugging Programs · Practise: 1.11 Tracing and Debugging Programs	907

Day	What to do	Minutes
2	Revise yesterday: How Java Works: Source, Bytecode, JVM, JRE, JDK, The Java Landscape: History, Features, Editions, JVM Memory, Installing Java on macOS and Choosing an Editor… · Learn: 2.1 One-Dimensional Arrays · Practise: 2.1 One-Dimensional Arrays · Learn: 2.2 Two-Dimensional Arrays (Matrices) · Practise: 2.2 Two-Dimensional Arrays (Matrices) · Learn: 2.3 Strings, StringBuilder and Characters · Practise: 2.3 Strings, StringBuilder and Characters · Learn: 2.4 Number Systems, Binary and Two's Complement · Practise: 2.4 Number Systems, Binary and Two's Complement · Learn: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic · Practise: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic · Learn: 2.6 Recursion: Base Case, Recursive Case and the Call Stack · Practise: 2.6 Recursion: Base Case, Recursive Case and the Call Stack · Learn: 2.7 References, Pass-by-Value and Memory · Practise: 2.7 References, Pass-by-Value and Memory · Learn: 2.8 Boundary Conditions, Validation and Common Mistakes · Practise: 2.8 Boundary Conditions, Validation and Common Mistakes · Learn: 3.1 Classes, Objects, Fields, Constructors and this · Practise: 3.1 Classes, Objects, Fields, Constructors and this · Learn: 3.2 Encapsulation and Access Modifiers · Practise: 3.2 Encapsulation and Access Modifiers · Learn: 3.3 Inheritance, Overriding and Polymorphism · Practise: 3.3 Inheritance, Overriding and Polymorphism · Learn: 3.4 Abstract Classes and Interfaces · Practise: 3.4 Abstract Classes and Interfaces · Learn: 3.5 static, final and Initialisation Order · Practise: 3.5 static, final and Initialisation Order · Learn: 3.6 Packages and Imports · Practise: 3.6 Packages and Imports	876
3	Revise yesterday: One-Dimensional Arrays, Two-Dimensional Arrays (Matrices), Strings, StringBuilder and Characters… · Learn: 3.7 Exceptions: try, catch, finally, throw and throws · Practise: 3.7 Exceptions: try, catch, finally, throw and throws · Learn: 3.8 Generics and Lists: ArrayList and LinkedList · Practise: 3.8 Generics and Lists: ArrayList and LinkedList · Learn: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Practise: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Learn: 3.10 Queues, Deques, Stacks and PriorityQueue · Practise: 3.10 Queues, Deques, Stacks and PriorityQueue · Learn: 3.11 Sorting with Comparable and Comparator · Practise: 3.11 Sorting with Comparable and Comparator · Learn: 3.12 equals(), hashCode() and Immutability · Practise: 3.12 equals(), hashCode() and Immutability · Learn: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Practise: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Learn: 4.2 Trees, Graphs and Heaps · Practise: 4.2 Trees, Graphs and Heaps · Learn: 4.3 Searching: Linear and Binary Search · Practise: 4.3 Searching: Linear and Binary Search · Learn: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability · Practise: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability · Learn: 4.5 Merge Sort and Divide and Conquer · Practise: 4.5 Merge Sort and Divide and Conquer · Learn: 4.6 Iteration versus Recursion · Practise: 4.6 Iteration versus Recursion · Learn: 4.7 Time Complexity and Big-O Notation · Practise: 4.7 Time Complexity and Big-O Notation	847
4	Revise yesterday: Exceptions: try, catch, finally, throw and throws, Generics and Lists: ArrayList and LinkedList, Sets and Maps: HashSet, TreeSet, HashMap, TreeMap… · Spaced review (3 days ago): How Java Works: Source, Bytecode, JVM, JRE, JDK, The Java Landscape: History, Features, Editions, JVM Memory · Learn: 4.8 Algorithm Design Paradigms · Practise: 4.8 Algorithm Design Paradigms · Learn: 4.9 Analysing Loops: Counting Iterations · Practise: 4.9 Analysing Loops: Counting Iterations · Learn: 4.10 Analysing Recursion: Recurrence Relations · Practise: 4.10 Analysing Recursion: Recurrence Relations · Learn: 4.11 Space Complexity and the Call Stack · Practise: 4.11 Space Complexity and the Call Stack · Learn: 4.12 Best, Average and Worst Case · Practise: 4.12 Best, Average and Worst Case · Learn: 5.1 Euclid's Algorithm (GCD) · Practise: 5.1 Euclid's Algorithm (GCD) · Learn: 5.2 Euler's Phi (Totient) Function · Practise: 5.2 Euler's Phi (Totient) Function · Learn: 5.3 Chinese Remainder Theorem · Practise: 5.3 Chinese Remainder Theorem · Learn: 5.4 Combinations (nCr) · Practise: 5.4 Combinations (nCr) · Learn: 5.5 Karatsuba's Multiplication · Practise: 5.5 Karatsuba's Multiplication · Learn: 5.6 Booth's Algorithm (signed binary multiplication) · Practise: 5.6 Booth's Algorithm (signed binary multiplication) · Learn: 5.7 Simple Sieve of Eratosthenes · Practise: 5.7 Simple Sieve of Eratosthenes · Learn: 5.8 Segmented & Incremental Sieve · Practise: 5.8 Segmented & Incremental Sieve	910

Day	What to do	Minutes
5	Revise yesterday: Algorithm Design Paradigms, Analysing Loops: Counting Iterations, Analysing Recursion: Recurrence Relations… · Spaced review (3 days ago): One-Dimensional Arrays, Two-Dimensional Arrays (Matrices) · Learn: 5.9 Binary Palindrome · Practise: 5.9 Binary Palindrome · Learn: 5.10 Swap Two Nibbles in a Byte · Practise: 5.10 Swap Two Nibbles in a Byte · Learn: 5.11 Longest Sequence of 1s (and after flipping a bit) · Practise: 5.11 Longest Sequence of 1s (and after flipping a bit) · Learn: 5.12 Josephus Problem · Practise: 5.12 Josephus Problem · Learn: 5.13 Block Swap Algorithm (Array Rotation) · Practise: 5.13 Block Swap Algorithm (Array Rotation) · Learn: 5.14 Leaders in an Array · Practise: 5.14 Leaders in an Array · Learn: 5.15 Majority Element (Boyer–Moore Voting) · Practise: 5.15 Majority Element (Boyer–Moore Voting) · Learn: 5.16 Maximum Product Subarray · Practise: 5.16 Maximum Product Subarray · Learn: 5.17 Maximum Equilibrium Sum · Practise: 5.17 Maximum Equilibrium Sum · Learn: 5.18 Maximum Sum of an Hourglass in a Matrix · Practise: 5.18 Maximum Sum of an Hourglass in a Matrix · Learn: 5.19 Toggle the Switch (100 Doors) · Practise: 5.19 Toggle the Switch (100 Doors) · Learn: 5.20 Alice's Apple Tree (Minimum Garden Perimeter) · Practise: 5.20 Alice's Apple Tree (Minimum Garden Perimeter) · Learn: 5.21 Weighted Substring (Smallest String of a Given Weight) · Practise: 5.21 Weighted Substring (Smallest String of a Given Weight) · Learn: 5.22 Move Hyphen to the Beginning · Practise: 5.22 Move Hyphen to the Beginning	924
6	Revise yesterday: Binary Palindrome, Swap Two Nibbles in a Byte, Longest Sequence of 1s (and after flipping a bit)… · Spaced review (3 days ago): Exceptions: try, catch, finally, throw and throws, Generics and Lists: ArrayList and LinkedList · Learn: 5.23 Strobogrammatic Number · Practise: 5.23 Strobogrammatic Number · Learn: 5.24 Lexicographically First Palindromic Rearrangement · Practise: 5.24 Lexicographically First Palindromic Rearrangement · Learn: 5.25 Manacher's Algorithm (Longest Palindromic Substring) · Practise: 5.25 Manacher's Algorithm (Longest Palindromic Substring) · Learn: 5.26 Natural Sort Order · Practise: 5.26 Natural Sort Order · Learn: 5.27 Maneuvering (Counting Grid Paths) · Practise: 5.27 Maneuvering (Counting Grid Paths) · Learn: 5.28 Sorted Distinct Permutations · Practise: 5.28 Sorted Distinct Permutations · Learn: 5.29 N-Queens (Backtracking) · Practise: 5.29 N-Queens (Backtracking) · Learn: 5.30 Maze Solving (Rat in a Maze) · Practise: 5.30 Maze Solving (Rat in a Maze) · Learn: 5.31 Hamiltonian Cycle · Practise: 5.31 Hamiltonian Cycle · Learn: 5.32 Knight's Tour & Warnsdorff's Rule · Practise: 5.32 Knight's Tour & Warnsdorff's Rule · Learn: 5.33 Selection Sort · Practise: 5.33 Selection Sort · Learn: 5.34 Quick Sort · Practise: 5.34 Quick Sort	901
7	Weak-topic review (from your quiz and practice results) · Final mixed assessment (archive topics)	75

14-day plan — minimum-viable path

Day	What to do	Minutes
1	Learn: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Practise: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Learn: 0.3 Installing Java on macOS and Choosing an Editor · Practise: 0.3 Installing Java on macOS and Choosing an Editor · Learn: 0.4 Your First Program: class, main, statements, comments · Practise: 0.4 Your First Program: class, main, statements, comments · Learn: 0.5 Compile-time, Runtime and Logic Errors · Practise: 0.5 Compile-time, Runtime and Logic Errors · Learn: 0.6 What Is an Algorithm? · Practise: 0.6 What Is an Algorithm? · Learn: 1.1 Variables, Constants and Primitive Types · Practise: 1.1 Variables, Constants and Primitive Types	289
2	Revise yesterday: How Java Works: Source, Bytecode, JVM, JRE, JDK, Installing Java on macOS and Choosing an Editor, Your First Program: class, main, statements, comments… · Learn: 1.2 Type Conversion, Casting and Overflow · Practise: 1.2 Type Conversion, Casting and Overflow · Learn: 1.3 Arithmetic, Assignment and Increment Operators · Practise: 1.3 Arithmetic, Assignment and Increment Operators · Learn: 1.4 Relational, Logical and Ternary Operators; Precedence · Practise: 1.4 Relational, Logical and Ternary Operators; Precedence · Learn: 1.5 Bitwise Operators and Shifts · Practise: 1.5 Bitwise Operators and Shifts	241
3	Revise yesterday: Type Conversion, Casting and Overflow, Arithmetic, Assignment and Increment Operators, Relational, Logical and Ternary Operators; Precedence… · Learn: 1.6 Output: print, println, printf and String Concatenation · Practise: 1.6 Output: print, println, printf and String Concatenation · Learn: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Practise: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Learn: 1.8 Decision Making: if, else, switch · Practise: 1.8 Decision Making: if, else, switch · Learn: 1.9 Loops: for, while, do-while, break and continue · Practise: 1.9 Loops: for, while, do-while, break and continue · Learn: 1.10 Methods: Parameters, Return Values, Scope and Overloading · Practise: 1.10 Methods: Parameters, Return Values, Scope and Overloading	310

Day	What to do	Minutes
4	Revise yesterday: Output: print, println, printf and String Concatenation, Input: Scanner, BufferedReader and Command-Line Arguments, Decision Making: if, else, switch... · Spaced review (3 days ago): How Java Works: Source, Bytecode, JVM, JRE, JDK, Installing Java on macOS and Choosing an Editor · Learn: 1.11 Tracing and Debugging Programs · Practise: 1.11 Tracing and Debugging Programs · Learn: 2.1 One-Dimensional Arrays · Practise: 2.1 One-Dimensional Arrays · Learn: 2.2 Two-Dimensional Arrays (Matrices) · Practise: 2.2 Two-Dimensional Arrays (Matrices) · Learn: 2.3 Strings, StringBuilder and Characters · Practise: 2.3 Strings, StringBuilder and Characters	269
5	Revise yesterday: Tracing and Debugging Programs, One-Dimensional Arrays, Two-Dimensional Arrays (Matrices)... · Spaced review (3 days ago): Type Conversion, Casting and Overflow, Arithmetic, Assignment and Increment Operators · Learn: 2.4 Number Systems, Binary and Two's Complement · Practise: 2.4 Number Systems, Binary and Two's Complement · Learn: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic · Practise: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic · Learn: 2.6 Recursion: Base Case, Recursive Case and the Call Stack · Practise: 2.6 Recursion: Base Case, Recursive Case and the Call Stack · Learn: 2.7 References, Pass-by-Value and Memory · Practise: 2.7 References, Pass-by-Value and Memory	284
6	Revise yesterday: Number Systems, Binary and Two's Complement, Divisibility, GCD/LCM, Primes and Modular Arithmetic, Recursion: Base Case, Recursive Case and the Call Stack... · Spaced review (3 days ago): Output: print, println, printf and String Concatenation, Input: Scanner, BufferedReader and Command-Line Arguments · Learn: 2.8 Boundary Conditions, Validation and Common Mistakes · Practise: 2.8 Boundary Conditions, Validation and Common Mistakes · Learn: 3.8 Generics and Lists: ArrayList and LinkedList · Practise: 3.8 Generics and Lists: ArrayList and LinkedList · Learn: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Practise: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Learn: 3.10 Queues, Deques, Stacks and PriorityQueue · Practise: 3.10 Queues, Deques, Stacks and PriorityQueue	259
7	Revise yesterday: Boundary Conditions, Validation and Common Mistakes, Generics and Lists: ArrayList and LinkedList, Sets and Maps: HashSet, TreeSet, HashMap, TreeMap... · Spaced review (3 days ago): Tracing and Debugging Programs, One-Dimensional Arrays · Learn: 3.11 Sorting with Comparable and Comparator · Practise: 3.11 Sorting with Comparable and Comparator · Learn: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Practise: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Learn: 4.3 Searching: Linear and Binary Search · Practise: 4.3 Searching: Linear and Binary Search · Learn: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability · Practise: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability	279
8	Revise yesterday: Sorting with Comparable and Comparator, Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables, Searching: Linear and Binary Search... · Spaced review (3 days ago): Number Systems, Binary and Two's Complement, Divisibility, GCD/LCM, Primes and Modular Arithmetic · Learn: 4.5 Merge Sort and Divide and Conquer · Practise: 4.5 Merge Sort and Divide and Conquer · Learn: 4.7 Time Complexity and Big-O Notation · Practise: 4.7 Time Complexity and Big-O Notation · Learn: 4.8 Algorithm Design Paradigms · Practise: 4.8 Algorithm Design Paradigms · Learn: 4.9 Analysing Loops: Counting Iterations · Practise: 4.9 Analysing Loops: Counting Iterations	304
9	Revise yesterday: Merge Sort and Divide and Conquer, Time Complexity and Big-O Notation, Algorithm Design Paradigms... · Spaced review (3 days ago): Boundary Conditions, Validation and Common Mistakes, Generics and Lists: ArrayList and LinkedList · Learn: 4.11 Space Complexity and the Call Stack · Practise: 4.11 Space Complexity and the Call Stack · Learn: 5.1 Euclid's Algorithm (GCD) · Practise: 5.1 Euclid's Algorithm (GCD) · Learn: 5.2 Euler's Phi (Totient) Function · Practise: 5.2 Euler's Phi (Totient) Function · Learn: 5.7 Simple Sieve of Eratosthenes · Practise: 5.7 Simple Sieve of Eratosthenes	269
10	Revise yesterday: Space Complexity and the Call Stack, Euclid's Algorithm (GCD), Euler's Phi (Totient) Function... · Spaced review (3 days ago): Sorting with Comparable and Comparator, Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Learn: 5.9 Binary Palindrome · Practise: 5.9 Binary Palindrome · Learn: 5.12 Josephus Problem · Practise: 5.12 Josephus Problem · Learn: 5.13 Block Swap Algorithm (Array Rotation) · Practise: 5.13 Block Swap Algorithm (Array Rotation) · Learn: 5.14 Leaders in an Array · Practise: 5.14 Leaders in an Array	279

Day	What to do	Minutes
11	Revise yesterday: Binary Palindrome, Josephus Problem, Block Swap Algorithm (Array Rotation)... · Spaced review (3 days ago): Merge Sort and Divide and Conquer, Time Complexity and Big-O Notation · Learn: 5.15 Majority Element (Boyer–Moore Voting) · Practise: 5.15 Majority Element (Boyer–Moore Voting) · Learn: 5.16 Maximum Product Subarray · Practise: 5.16 Maximum Product Subarray · Learn: 5.17 Maximum Equilibrium Sum · Practise: 5.17 Maximum Equilibrium Sum · Learn: 5.18 Maximum Sum of an Hourglass in a Matrix · Practise: 5.18 Maximum Sum of an Hourglass in a Matrix	279
12	Revise yesterday: Majority Element (Boyer–Moore Voting), Maximum Product Subarray, Maximum Equilibrium Sum... · Spaced review (3 days ago): Space Complexity and the Call Stack, Euclid's Algorithm (GCD) · Learn: 5.19 Toggle the Switch (100 Doors) · Practise: 5.19 Toggle the Switch (100 Doors) · Learn: 5.22 Move Hyphen to the Beginning · Practise: 5.22 Move Hyphen to the Beginning · Learn: 5.24 Lexicographically First Palindromic Rearrangement · Practise: 5.24 Lexicographically First Palindromic Rearrangement · Learn: 5.27 Maneuvering (Counting Grid Paths) · Practise: 5.27 Maneuvering (Counting Grid Paths)	269
13	Revise yesterday: Toggle the Switch (100 Doors), Move Hyphen to the Beginning, Lexicographically First Palindromic Rearrangement... · Spaced review (3 days ago): Binary Palindrome, Josephus Problem · Learn: 5.29 N-Queens (Backtracking) · Practise: 5.29 N-Queens (Backtracking) · Learn: 5.30 Maze Solving (Rat in a Maze) · Practise: 5.30 Maze Solving (Rat in a Maze) · Learn: 5.33 Selection Sort · Practise: 5.33 Selection Sort · Learn: 5.34 Quick Sort · Practise: 5.34 Quick Sort	319
14	Weak-topic review (from your quiz and practice results) · Final mixed assessment (archive topics)	75

14-day plan — complete path

Day	What to do	Minutes
1	Learn: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Practise: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Learn: 0.2 The Java Landscape: History, Features, Editions, JVM Memory · Practise: 0.2 The Java Landscape: History, Features, Editions, JVM Memory · Learn: 0.3 Installing Java on macOS and Choosing an Editor · Practise: 0.3 Installing Java on macOS and Choosing an Editor · Learn: 0.4 Your First Program: class, main, statements, comments · Practise: 0.4 Your First Program: class, main, statements, comments · Learn: 0.5 Compile-time, Runtime and Logic Errors · Practise: 0.5 Compile-time, Runtime and Logic Errors · Learn: 0.6 What Is an Algorithm? · Practise: 0.6 What Is an Algorithm? · Learn: 1.1 Variables, Constants and Primitive Types · Practise: 1.1 Variables, Constants and Primitive Types · Learn: 1.2 Type Conversion, Casting and Overflow · Practise: 1.2 Type Conversion, Casting and Overflow	381
2	Revise yesterday: How Java Works: Source, Bytecode, JVM, JRE, JDK, The Java Landscape: History, Features, Editions, JVM Memory, Installing Java on macOS and Choosing an Editor... · Learn: 1.3 Arithmetic, Assignment and Increment Operators · Practise: 1.3 Arithmetic, Assignment and Increment Operators · Learn: 1.4 Relational, Logical and Ternary Operators; Precedence · Practise: 1.4 Relational, Logical and Ternary Operators; Precedence · Learn: 1.5 Bitwise Operators and Shifts · Practise: 1.5 Bitwise Operators and Shifts · Learn: 1.6 Output: print, println, printf and String Concatenation · Practise: 1.6 Output: print, println, printf and String Concatenation · Learn: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Practise: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Learn: 1.8 Decision Making: if, else, switch · Practise: 1.8 Decision Making: if, else, switch · Learn: 1.9 Loops: for, while, do-while, break and continue · Practise: 1.9 Loops: for, while, do-while, break and continue	418
3	Revise yesterday: Arithmetic, Assignment and Increment Operators, Relational, Logical and Ternary Operators; Precedence, Bitwise Operators and Shifts... · Learn: 1.10 Methods: Parameters, Return Values, Scope and Overloading · Practise: 1.10 Methods: Parameters, Return Values, Scope and Overloading · Learn: 1.11 Tracing and Debugging Programs · Practise: 1.11 Tracing and Debugging Programs · Learn: 2.1 One-Dimensional Arrays · Practise: 2.1 One-Dimensional Arrays · Learn: 2.2 Two-Dimensional Arrays (Matrices) · Practise: 2.2 Two-Dimensional Arrays (Matrices) · Learn: 2.3 Strings, StringBuilder and Characters · Practise: 2.3 Strings, StringBuilder and Characters · Learn: 2.4 Number Systems, Binary and Two's Complement · Practise: 2.4 Number Systems, Binary and Two's Complement · Learn: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic · Practise: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic	458

Day	What to do	Minutes
4	Revise yesterday: Methods: Parameters, Return Values, Scope and Overloading, Tracing and Debugging Programs, One-Dimensional Arrays… · Spaced review (3 days ago): How Java Works: Source, Bytecode, JVM, JRE, JDK, The Java Landscape: History, Features, Editions, JVM Memory · Learn: 2.6 Recursion: Base Case, Recursive Case and the Call Stack · Practise: 2.6 Recursion: Base Case, Recursive Case and the Call Stack · Learn: 2.7 References, Pass-by-Value and Memory · Practise: 2.7 References, Pass-by-Value and Memory · Learn: 2.8 Boundary Conditions, Validation and Common Mistakes · Practise: 2.8 Boundary Conditions, Validation and Common Mistakes · Learn: 3.1 Classes, Objects, Fields, Constructors and this · Practise: 3.1 Classes, Objects, Fields, Constructors and this · Learn: 3.2 Encapsulation and Access Modifiers · Practise: 3.2 Encapsulation and Access Modifiers · Learn: 3.3 Inheritance, Overriding and Polymorphism · Practise: 3.3 Inheritance, Overriding and Polymorphism	392
5	Revise yesterday: Recursion: Base Case, Recursive Case and the Call Stack, References, Pass-by-Value and Memory, Boundary Conditions, Validation and Common Mistakes… · Spaced review (3 days ago): Arithmetic, Assignment and Increment Operators, Relational, Logical and Ternary Operators; Precedence · Learn: 3.4 Abstract Classes and Interfaces · Practise: 3.4 Abstract Classes and Interfaces · Learn: 3.5 static, final and Initialisation Order · Practise: 3.5 static, final and Initialisation Order · Learn: 3.6 Packages and Imports · Practise: 3.6 Packages and Imports · Learn: 3.7 Exceptions: try, catch, finally, throw and throws · Practise: 3.7 Exceptions: try, catch, finally, throw and throws · Learn: 3.8 Generics and Lists: ArrayList and LinkedList · Practise: 3.8 Generics and Lists: ArrayList and LinkedList · Learn: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Practise: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Learn: 3.10 Queues, Deques, Stacks and PriorityQueue · Practise: 3.10 Queues, Deques, Stacks and PriorityQueue	426
6	Revise yesterday: Abstract Classes and Interfaces, static, final and Initialisation Order, Packages and Imports… · Spaced review (3 days ago): Methods: Parameters, Return Values, Scope and Overloading, Tracing and Debugging Programs · Learn: 3.11 Sorting with Comparable and Comparator · Practise: 3.11 Sorting with Comparable and Comparator · Learn: 3.12 equals(), hashCode() and Immutability · Practise: 3.12 equals(), hashCode() and Immutability · Learn: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Practise: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Learn: 4.2 Trees, Graphs and Heaps · Practise: 4.2 Trees, Graphs and Heaps · Learn: 4.3 Searching: Linear and Binary Search · Practise: 4.3 Searching: Linear and Binary Search · Learn: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability · Practise: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability	412
7	Revise yesterday: Sorting with Comparable and Comparator, equals(), hashCode() and Immutability, Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables… · Spaced review (3 days ago): Recursion: Base Case, Recursive Case and the Call Stack, References, Pass-by-Value and Memory · Learn: 4.5 Merge Sort and Divide and Conquer · Practise: 4.5 Merge Sort and Divide and Conquer · Learn: 4.6 Iteration versus Recursion · Practise: 4.6 Iteration versus Recursion · Learn: 4.7 Time Complexity and Big-O Notation · Practise: 4.7 Time Complexity and Big-O Notation · Learn: 4.8 Algorithm Design Paradigms · Practise: 4.8 Algorithm Design Paradigms · Learn: 4.9 Analysing Loops: Counting Iterations · Practise: 4.9 Analysing Loops: Counting Iterations · Learn: 4.10 Analysing Recursion: Recurrence Relations · Practise: 4.10 Analysing Recursion: Recurrence Relations · Learn: 4.11 Space Complexity and the Call Stack · Practise: 4.11 Space Complexity and the Call Stack	481
8	Revise yesterday: Merge Sort and Divide and Conquer, Iteration versus Recursion, Time Complexity and Big-O Notation… · Spaced review (3 days ago): Abstract Classes and Interfaces, static, final and Initialisation Order · Learn: 4.12 Best, Average and Worst Case · Practise: 4.12 Best, Average and Worst Case · Learn: 5.1 Euclid's Algorithm (GCD) · Practise: 5.1 Euclid's Algorithm (GCD) · Learn: 5.2 Euler's Phi (Totient) Function · Practise: 5.2 Euler's Phi (Totient) Function · Learn: 5.3 Chinese Remainder Theorem · Practise: 5.3 Chinese Remainder Theorem · Learn: 5.4 Combinations (nCr) · Practise: 5.4 Combinations (nCr) · Learn: 5.5 Karatsuba's Multiplication · Practise: 5.5 Karatsuba's Multiplication	417
9	Revise yesterday: Best, Average and Worst Case, Euclid's Algorithm (GCD), Euler's Phi (Totient) Function… · Spaced review (3 days ago): Sorting with Comparable and Comparator, equals(), hashCode() and Immutability · Learn: 5.6 Booth's Algorithm (signed binary multiplication) · Practise: 5.6 Booth's Algorithm (signed binary multiplication) · Learn: 5.7 Simple Sieve of Eratosthenes · Practise: 5.7 Simple Sieve of Eratosthenes · Learn: 5.8 Segmented & Incremental Sieve · Practise: 5.8 Segmented & Incremental Sieve · Learn: 5.9 Binary Palindrome · Practise: 5.9 Binary Palindrome · Learn: 5.10 Swap Two Nibbles in a Byte · Practise: 5.10 Swap Two Nibbles in a Byte · Learn: 5.11 Longest Sequence of 1s (and after flipping a bit) · Practise: 5.11 Longest Sequence of 1s (and after flipping a bit)	427

Day	What to do	Minutes
10	Revise yesterday: Booth's Algorithm (signed binary multiplication), Simple Sieve of Eratosthenes, Segmented & Incremental Sieve… · Spaced review (3 days ago): Merge Sort and Divide and Conquer, Iteration versus Recursion · Learn: 5.12 Josephus Problem · Practise: 5.12 Josephus Problem · Learn: 5.13 Block Swap Algorithm (Array Rotation) · Practise: 5.13 Block Swap Algorithm (Array Rotation) · Learn: 5.14 Leaders in an Array · Practise: 5.14 Leaders in an Array · Learn: 5.15 Majority Element (Boyer–Moore Voting) · Practise: 5.15 Majority Element (Boyer–Moore Voting) · Learn: 5.16 Maximum Product Subarray · Practise: 5.16 Maximum Product Subarray · Learn: 5.17 Maximum Equilibrium Sum · Practise: 5.17 Maximum Equilibrium Sum	427
11	Revise yesterday: Josephus Problem, Block Swap Algorithm (Array Rotation), Leaders in an Array… · Spaced review (3 days ago): Best, Average and Worst Case, Euclid's Algorithm (GCD) · Learn: 5.18 Maximum Sum of an Hourglass in a Matrix · Practise: 5.18 Maximum Sum of an Hourglass in a Matrix · Learn: 5.19 Toggle the Switch (100 Doors) · Practise: 5.19 Toggle the Switch (100 Doors) · Learn: 5.20 Alice's Apple Tree (Minimum Garden Perimeter) · Practise: 5.20 Alice's Apple Tree (Minimum Garden Perimeter) · Learn: 5.21 Weighted Substring (Smallest String of a Given Weight) · Practise: 5.21 Weighted Substring (Smallest String of a Given Weight) · Learn: 5.22 Move Hyphen to the Beginning · Practise: 5.22 Move Hyphen to the Beginning · Learn: 5.23 Strobogrammatic Number · Practise: 5.23 Strobogrammatic Number	397
12	Revise yesterday: Maximum Sum of an Hourglass in a Matrix, Toggle the Switch (100 Doors), Alice's Apple Tree (Minimum Garden Perimeter)… · Spaced review (3 days ago): Booth's Algorithm (signed binary multiplication), Simple Sieve of Eratosthenes · Learn: 5.24 Lexicographically First Palindromic Rearrangement · Practise: 5.24 Lexicographically First Palindromic Rearrangement · Learn: 5.25 Manacher's Algorithm (Longest Palindromic Substring) · Practise: 5.25 Manacher's Algorithm (Longest Palindromic Substring) · Learn: 5.26 Natural Sort Order · Practise: 5.26 Natural Sort Order · Learn: 5.27 Maneuvering (Counting Grid Paths) · Practise: 5.27 Maneuvering (Counting Grid Paths) · Learn: 5.28 Sorted Distinct Permutations · Practise: 5.28 Sorted Distinct Permutations · Learn: 5.29 N-Queens (Backtracking) · Practise: 5.29 N-Queens (Backtracking)	462
13	Revise yesterday: Lexicographically First Palindromic Rearrangement, Manacher's Algorithm (Longest Palindromic Substring), Natural Sort Order… · Spaced review (3 days ago): Josephus Problem, Block Swap Algorithm (Array Rotation) · Learn: 5.30 Maze Solving (Rat in a Maze) · Practise: 5.30 Maze Solving (Rat in a Maze) · Learn: 5.31 Hamiltonian Cycle · Practise: 5.31 Hamiltonian Cycle · Learn: 5.32 Knight's Tour & Warnsdorff's Rule · Practise: 5.32 Knight's Tour & Warnsdorff's Rule · Learn: 5.33 Selection Sort · Practise: 5.33 Selection Sort · Learn: 5.34 Quick Sort · Practise: 5.34 Quick Sort	393
14	Weak-topic review (from your quiz and practice results) · Final mixed assessment (archive topics)	75

21-day plan — minimum-viable path

Day	What to do	Minutes
1	Learn: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Practise: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Learn: 0.3 Installing Java on macOS and Choosing an Editor · Practise: 0.3 Installing Java on macOS and Choosing an Editor · Learn: 0.4 Your First Program: class, main, statements, comments · Practise: 0.4 Your First Program: class, main, statements, comments · Learn: 0.5 Compile-time, Runtime and Logic Errors · Practise: 0.5 Compile-time, Runtime and Logic Errors	176
2	Revise yesterday: How Java Works: Source, Bytecode, JVM, JRE, JDK, Installing Java on macOS and Choosing an Editor, Your First Program: class, main, statements, comments… · Learn: 0.6 What Is an Algorithm? · Practise: 0.6 What Is an Algorithm? · Learn: 1.1 Variables, Constants and Primitive Types · Practise: 1.1 Variables, Constants and Primitive Types · Learn: 1.2 Type Conversion, Casting and Overflow · Practise: 1.2 Type Conversion, Casting and Overflow	182
3	Revise yesterday: What Is an Algorithm?, Variables, Constants and Primitive Types, Type Conversion, Casting and Overflow · Learn: 1.3 Arithmetic, Assignment and Increment Operators · Practise: 1.3 Arithmetic, Assignment and Increment Operators · Learn: 1.4 Relational, Logical and Ternary Operators; Precedence · Practise: 1.4 Relational, Logical and Ternary Operators; Precedence · Learn: 1.5 Bitwise Operators and Shifts · Practise: 1.5 Bitwise Operators and Shifts	182

Day	What to do	Minutes
4	Revise yesterday: Arithmetic, Assignment and Increment Operators, Relational, Logical and Ternary Operators; Precedence, Bitwise Operators and Shifts · Spaced review (3 days ago): How Java Works: Source, Bytecode, JVM, JRE, JDK, Installing Java on macOS and Choosing an Editor · Learn: 1.6 Output: print, println, printf and String Concatenation · Practise: 1.6 Output: print, println, printf and String Concatenation · Learn: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Practise: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Learn: 1.8 Decision Making: if, else, switch · Practise: 1.8 Decision Making: if, else, switch	190
5	Revise yesterday: Output: print, println, printf and String Concatenation, Input: Scanner, BufferedReader and Command-Line Arguments, Decision Making: if, else, switch · Spaced review (3 days ago): What Is an Algorithm?, Variables, Constants and Primitive Types · Learn: 1.9 Loops: for, while, do-while, break and continue · Practise: 1.9 Loops: for, while, do-while, break and continue · Learn: 1.10 Methods: Parameters, Return Values, Scope and Overloading · Practise: 1.10 Methods: Parameters, Return Values, Scope and Overloading · Learn: 1.11 Tracing and Debugging Programs · Practise: 1.11 Tracing and Debugging Programs	200
6	Revise yesterday: Loops: for, while, do-while, break and continue, Methods: Parameters, Return Values, Scope and Overloading, Tracing and Debugging Programs · Spaced review (3 days ago): Arithmetic, Assignment and Increment Operators, Relational, Logical and Ternary Operators; Precedence · Learn: 2.1 One-Dimensional Arrays · Practise: 2.1 One-Dimensional Arrays · Learn: 2.2 Two-Dimensional Arrays (Matrices) · Practise: 2.2 Two-Dimensional Arrays (Matrices)	146
7	Revise yesterday: One-Dimensional Arrays, Two-Dimensional Arrays (Matrices) · Spaced review (3 days ago): Output: print, println, printf and String Concatenation, Input: Scanner, BufferedReader and Command-Line Arguments · Learn: 2.3 Strings, StringBuilder and Characters · Practise: 2.3 Strings, StringBuilder and Characters · Learn: 2.4 Number Systems, Binary and Two's Complement · Practise: 2.4 Number Systems, Binary and Two's Complement · Learn: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic · Practise: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic	220
8	Revise yesterday: Strings, StringBuilder and Characters, Number Systems, Binary and Two's Complement, Divisibility, GCD/LCM, Primes and Modular Arithmetic · Spaced review (3 days ago): Loops: for, while, do-while, break and continue, Methods: Parameters, Return Values, Scope and Overloading · Learn: 2.6 Recursion: Base Case, Recursive Case and the Call Stack · Practise: 2.6 Recursion: Base Case, Recursive Case and the Call Stack · Learn: 2.7 References, Pass-by-Value and Memory · Practise: 2.7 References, Pass-by-Value and Memory · Learn: 2.8 Boundary Conditions, Validation and Common Mistakes · Practise: 2.8 Boundary Conditions, Validation and Common Mistakes	205
9	Revise yesterday: Recursion: Base Case, Recursive Case and the Call Stack, References, Pass-by-Value and Memory, Boundary Conditions, Validation and Common Mistakes · Spaced review (3 days ago): One-Dimensional Arrays, Two-Dimensional Arrays (Matrices) · Learn: 3.8 Generics and Lists: ArrayList and LinkedList · Practise: 3.8 Generics and Lists: ArrayList and LinkedList · Learn: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Practise: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Learn: 3.10 Queues, Deques, Stacks and PriorityQueue · Practise: 3.10 Queues, Deques, Stacks and PriorityQueue	205
10	Revise yesterday: Generics and Lists: ArrayList and LinkedList, Sets and Maps: HashSet, TreeSet, HashMap, TreeMap, Queues, Deques, Stacks and PriorityQueue · Spaced review (3 days ago): Strings, StringBuilder and Characters, Number Systems, Binary and Two's Complement · Learn: 3.11 Sorting with Comparable and Comparator · Practise: 3.11 Sorting with Comparable and Comparator · Learn: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Practise: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables	146
11	Revise yesterday: Sorting with Comparable and Comparator, Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Spaced review (3 days ago): Recursion: Base Case, Recursive Case and the Call Stack, References, Pass-by-Value and Memory · Learn: 4.3 Searching: Linear and Binary Search · Practise: 4.3 Searching: Linear and Binary Search · Learn: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability · Practise: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability · Learn: 4.5 Merge Sort and Divide and Conquer · Practise: 4.5 Merge Sort and Divide and Conquer	220

Day	What to do	Minutes
12	Revise yesterday: Searching: Linear and Binary Search, Elementary Sorting: Bubble, Insertion, Selection; Stability, Merge Sort and Divide and Conquer · Spaced review (3 days ago): Generics and Lists: ArrayList and LinkedList, Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Learn: 4.7 Time Complexity and Big-O Notation · Practise: 4.7 Time Complexity and Big-O Notation · Learn: 4.8 Algorithm Design Paradigms · Practise: 4.8 Algorithm Design Paradigms	161
13	Revise yesterday: Time Complexity and Big-O Notation, Algorithm Design Paradigms · Spaced review (3 days ago): Sorting with Comparable and Comparator, Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Learn: 4.9 Analysing Loops: Counting Iterations · Practise: 4.9 Analysing Loops: Counting Iterations · Learn: 4.11 Space Complexity and the Call Stack · Practise: 4.11 Space Complexity and the Call Stack · Learn: 5.1 Euclid's Algorithm (GCD) · Practise: 5.1 Euclid's Algorithm (GCD)	210
14	Revise yesterday: Analysing Loops: Counting Iterations, Space Complexity and the Call Stack, Euclid's Algorithm (GCD) · Spaced review (3 days ago): Searching: Linear and Binary Search, Elementary Sorting: Bubble, Insertion, Selection; Stability · Learn: 5.2 Euler's Phi (Totient) Function · Practise: 5.2 Euler's Phi (Totient) Function · Learn: 5.7 Simple Sieve of Eratosthenes · Practise: 5.7 Simple Sieve of Eratosthenes · Learn: 5.9 Binary Palindrome · Practise: 5.9 Binary Palindrome	210
15	Revise yesterday: Euler's Phi (Totient) Function, Simple Sieve of Eratosthenes, Binary Palindrome · Spaced review (3 days ago): Time Complexity and Big-O Notation, Algorithm Design Paradigms · Learn: 5.12 Josephus Problem · Practise: 5.12 Josephus Problem · Learn: 5.13 Block Swap Algorithm (Array Rotation) · Practise: 5.13 Block Swap Algorithm (Array Rotation)	166
16	Revise yesterday: Josephus Problem, Block Swap Algorithm (Array Rotation) · Spaced review (3 days ago): Analysing Loops: Counting Iterations, Space Complexity and the Call Stack · Learn: 5.14 Leaders in an Array · Practise: 5.14 Leaders in an Array · Learn: 5.15 Majority Element (Boyer–Moore Voting) · Practise: 5.15 Majority Element (Boyer–Moore Voting) · Learn: 5.16 Maximum Product Subarray · Practise: 5.16 Maximum Product Subarray	220
17	Revise yesterday: Leaders in an Array, Majority Element (Boyer–Moore Voting), Maximum Product Subarray · Spaced review (3 days ago): Euler's Phi (Totient) Function, Simple Sieve of Eratosthenes · Learn: 5.17 Maximum Equilibrium Sum · Practise: 5.17 Maximum Equilibrium Sum · Learn: 5.18 Maximum Sum of an Hourglass in a Matrix · Practise: 5.18 Maximum Sum of an Hourglass in a Matrix · Learn: 5.19 Toggle the Switch (100 Doors) · Practise: 5.19 Toggle the Switch (100 Doors)	195
18	Revise yesterday: Maximum Equilibrium Sum, Maximum Sum of an Hourglass in a Matrix, Toggle the Switch (100 Doors) · Spaced review (3 days ago): Josephus Problem, Block Swap Algorithm (Array Rotation) · Learn: 5.22 Move Hyphen to the Beginning · Practise: 5.22 Move Hyphen to the Beginning · Learn: 5.24 Lexicographically First Palindromic Rearrangement · Practise: 5.24 Lexicographically First Palindromic Rearrangement	136
19	Revise yesterday: Move Hyphen to the Beginning, Lexicographically First Palindromic Rearrangement · Spaced review (3 days ago): Leaders in an Array, Majority Element (Boyer–Moore Voting) · Learn: 5.27 Maneuvering (Counting Grid Paths) · Practise: 5.27 Maneuvering (Counting Grid Paths) · Learn: 5.29 N-Queens (Backtracking) · Practise: 5.29 N-Queens (Backtracking) · Learn: 5.30 Maze Solving (Rat in a Maze) · Practise: 5.30 Maze Solving (Rat in a Maze)	250
20	Revise yesterday: Maneuvering (Counting Grid Paths), N-Queens (Backtracking), Maze Solving (Rat in a Maze) · Spaced review (3 days ago): Maximum Equilibrium Sum, Maximum Sum of an Hourglass in a Matrix · Learn: 5.33 Selection Sort · Practise: 5.33 Selection Sort · Learn: 5.34 Quick Sort · Practise: 5.34 Quick Sort	156
21	Weak-topic review (from your quiz and practice results) · Final mixed assessment (archive topics)	75

21-day plan — complete path

Day	What to do	Minutes
1	Learn: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Practise: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Learn: 0.2 The Java Landscape: History, Features, Editions, JVM Memory · Practise: 0.2 The Java Landscape: History, Features, Editions, JVM Memory · Learn: 0.3 Installing Java on macOS and Choosing an Editor · Practise: 0.3 Installing Java on macOS and Choosing an Editor · Learn: 0.4 Your First Program: class, main, statements, comments · Practise: 0.4 Your First Program: class, main, statements, comments · Learn: 0.5 Compile-time, Runtime and Logic Errors · Practise: 0.5 Compile-time, Runtime and Logic Errors · Learn: 0.6 What Is an Algorithm? · Practise: 0.6 What Is an Algorithm?	263
2	Revise yesterday: How Java Works: Source, Bytecode, JVM, JRE, JDK, The Java Landscape: History, Features, Editions, JVM Memory, Installing Java on macOS and Choosing an Editor… · Learn: 1.1 Variables, Constants and Primitive Types · Practise: 1.1 Variables, Constants and Primitive Types · Learn: 1.2 Type Conversion, Casting and Overflow · Practise: 1.2 Type Conversion, Casting and Overflow · Learn: 1.3 Arithmetic, Assignment and Increment Operators · Practise: 1.3 Arithmetic, Assignment and Increment Operators · Learn: 1.4 Relational, Logical and Ternary Operators; Precedence · Practise: 1.4 Relational, Logical and Ternary Operators; Precedence · Learn: 1.5 Bitwise Operators and Shifts · Practise: 1.5 Bitwise Operators and Shifts	300
3	Revise yesterday: Variables, Constants and Primitive Types, Type Conversion, Casting and Overflow, Arithmetic, Assignment and Increment Operators… · Learn: 1.6 Output: print, println, printf and String Concatenation · Practise: 1.6 Output: print, println, printf and String Concatenation · Learn: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Practise: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Learn: 1.8 Decision Making: if, else, switch · Practise: 1.8 Decision Making: if, else, switch · Learn: 1.9 Loops: for, while, do-while, break and continue · Practise: 1.9 Loops: for, while, do-while, break and continue	246
4	Revise yesterday: Output: print, println, printf and String Concatenation, Input: Scanner, BufferedReader and Command-Line Arguments, Decision Making: if, else, switch… · Spaced review (3 days ago): How Java Works: Source, Bytecode, JVM, JRE, JDK, The Java Landscape: History, Features, Editions, JVM Memory · Learn: 1.10 Methods: Parameters, Return Values, Scope and Overloading · Practise: 1.10 Methods: Parameters, Return Values, Scope and Overloading · Learn: 1.11 Tracing and Debugging Programs · Practise: 1.11 Tracing and Debugging Programs · Learn: 2.1 One-Dimensional Arrays · Practise: 2.1 One-Dimensional Arrays · Learn: 2.2 Two-Dimensional Arrays (Matrices) · Practise: 2.2 Two-Dimensional Arrays (Matrices)	264
5	Revise yesterday: Methods: Parameters, Return Values, Scope and Overloading, Tracing and Debugging Programs, One-Dimensional Arrays… · Spaced review (3 days ago): Variables, Constants and Primitive Types, Type Conversion, Casting and Overflow · Learn: 2.3 Strings, StringBuilder and Characters · Practise: 2.3 Strings, StringBuilder and Characters · Learn: 2.4 Number Systems, Binary and Two's Complement · Practise: 2.4 Number Systems, Binary and Two's Complement · Learn: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic · Practise: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic · Learn: 2.6 Recursion: Base Case, Recursive Case and the Call Stack · Practise: 2.6 Recursion: Base Case, Recursive Case and the Call Stack	294
6	Revise yesterday: Strings, StringBuilder and Characters, Number Systems, Binary and Two's Complement, Divisibility, GCD/LCM, Primes and Modular Arithmetic… · Spaced review (3 days ago): Output: print, println, printf and String Concatenation, Input: Scanner, BufferedReader and Command-Line Arguments · Learn: 2.7 References, Pass-by-Value and Memory · Practise: 2.7 References, Pass-by-Value and Memory · Learn: 2.8 Boundary Conditions, Validation and Common Mistakes · Practise: 2.8 Boundary Conditions, Validation and Common Mistakes · Learn: 3.1 Classes, Objects, Fields, Constructors and this · Practise: 3.1 Classes, Objects, Fields, Constructors and this · Learn: 3.2 Encapsulation and Access Modifiers · Practise: 3.2 Encapsulation and Access Modifiers · Learn: 3.3 Inheritance, Overriding and Polymorphism · Practise: 3.3 Inheritance, Overriding and Polymorphism	318

Day	What to do	Minutes
7	Revise yesterday: References, Pass-by-Value and Memory, Boundary Conditions, Validation and Common Mistakes, Classes, Objects, Fields, Constructors and `this`... · Spaced review (3 days ago): Methods: Parameters, Return Values, Scope and Overloading, Tracing and Debugging Programs · Learn: 3.4 Abstract Classes and Interfaces · Practise: 3.4 Abstract Classes and Interfaces · Learn: 3.5 static, final and Initialisation Order · Practise: 3.5 static, final and Initialisation Order · Learn: 3.6 Packages and Imports · Practise: 3.6 Packages and Imports · Learn: 3.7 Exceptions: try, catch, finally, throw and throws · Practise: 3.7 Exceptions: try, catch, finally, throw and throws	239
8	Revise yesterday: Abstract Classes and Interfaces, static, final and Initialisation Order, Packages and Imports... · Spaced review (3 days ago): Strings, StringBuilder and Characters, Number Systems, Binary and Two's Complement · Learn: 3.8 Generics and Lists: ArrayList and LinkedList · Practise: 3.8 Generics and Lists: ArrayList and LinkedList · Learn: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Practise: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Learn: 3.10 Queues, Deques, Stacks and PriorityQueue · Practise: 3.10 Queues, Deques, Stacks and PriorityQueue · Learn: 3.11 Sorting with Comparable and Comparator · Practise: 3.11 Sorting with Comparable and Comparator · Learn: 3.12 equals(), hashCode() and Immutability · Practise: 3.12 equals(), hashCode() and Immutability	323
9	Revise yesterday: Generics and Lists: ArrayList and LinkedList, Sets and Maps: HashSet, TreeSet, HashMap, TreeMap, Queues, Deques, Stacks and PriorityQueue... · Spaced review (3 days ago): References, Pass-by-Value and Memory, Boundary Conditions, Validation and Common Mistakes · Learn: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Practise: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Learn: 4.2 Trees, Graphs and Heaps · Practise: 4.2 Trees, Graphs and Heaps · Learn: 4.3 Searching: Linear and Binary Search · Practise: 4.3 Searching: Linear and Binary Search · Learn: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability · Practise: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability	294
10	Revise yesterday: Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables, Trees, Graphs and Heaps, Searching: Linear and Binary Search... · Spaced review (3 days ago): Abstract Classes and Interfaces, static, final and Initialisation Order · Learn: 4.5 Merge Sort and Divide and Conquer · Practise: 4.5 Merge Sort and Divide and Conquer · Learn: 4.6 Iteration versus Recursion · Practise: 4.6 Iteration versus Recursion · Learn: 4.7 Time Complexity and Big-O Notation · Practise: 4.7 Time Complexity and Big-O Notation	215
11	Revise yesterday: Merge Sort and Divide and Conquer, Iteration versus Recursion, Time Complexity and Big-O Notation · Spaced review (3 days ago): Generics and Lists: ArrayList and LinkedList, Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Learn: 4.8 Algorithm Design Paradigms · Practise: 4.8 Algorithm Design Paradigms · Learn: 4.9 Analysing Loops: Counting Iterations · Practise: 4.9 Analysing Loops: Counting Iterations · Learn: 4.10 Analysing Recursion: Recurrence Relations · Practise: 4.10 Analysing Recursion: Recurrence Relations · Learn: 4.11 Space Complexity and the Call Stack · Practise: 4.11 Space Complexity and the Call Stack · Learn: 4.12 Best, Average and Worst Case · Practise: 4.12 Best, Average and Worst Case	338
12	Revise yesterday: Algorithm Design Paradigms, Analysing Loops: Counting Iterations, Analysing Recursion: Recurrence Relations... · Spaced review (3 days ago): Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables, Trees, Graphs and Heaps · Learn: 5.1 Euclid's Algorithm (GCD) · Practise: 5.1 Euclid's Algorithm (GCD) · Learn: 5.2 Euler's Phi (Totient) Function · Practise: 5.2 Euler's Phi (Totient) Function · Learn: 5.3 Chinese Remainder Theorem · Practise: 5.3 Chinese Remainder Theorem · Learn: 5.4 Combinations (nCr) · Practise: 5.4 Combinations (nCr)	284
13	Revise yesterday: Euclid's Algorithm (GCD), Euler's Phi (Totient) Function, Chinese Remainder Theorem... · Spaced review (3 days ago): Merge Sort and Divide and Conquer, Iteration versus Recursion · Learn: 5.5 Karatsuba's Multiplication · Practise: 5.5 Karatsuba's Multiplication · Learn: 5.6 Booth's Algorithm (signed binary multiplication) · Practise: 5.6 Booth's Algorithm (signed binary multiplication) · Learn: 5.7 Simple Sieve of Eratosthenes · Practise: 5.7 Simple Sieve of Eratosthenes	245
14	Revise yesterday: Karatsuba's Multiplication, Booth's Algorithm (signed binary multiplication), Simple Sieve of Eratosthenes · Spaced review (3 days ago): Algorithm Design Paradigms, Analysing Loops: Counting Iterations · Learn: 5.8 Segmented & Incremental Sieve · Practise: 5.8 Segmented & Incremental Sieve · Learn: 5.9 Binary Palindrome · Practise: 5.9 Binary Palindrome · Learn: 5.10 Swap Two Nibbles in a Byte · Practise: 5.10 Swap Two Nibbles in a Byte · Learn: 5.11 Longest Sequence of 1s (and after flipping a bit) · Practise: 5.11 Longest Sequence of 1s (and after flipping a bit)	279

Day	What to do	Minutes
15	Revise yesterday: Segmented & Incremental Sieve, Binary Palindrome, Swap Two Nibbles in a Byte... · Spaced review (3 days ago): Euclid's Algorithm (GCD), Euler's Phi (Totient) Function · Learn: 5.12 Josephus Problem · Practise: 5.12 Josephus Problem · Learn: 5.13 Block Swap Algorithm (Array Rotation) · Practise: 5.13 Block Swap Algorithm (Array Rotation) · Learn: 5.14 Leaders in an Array · Practise: 5.14 Leaders in an Array · Learn: 5.15 Majority Element (Boyer–Moore Voting) · Practise: 5.15 Majority Element (Boyer–Moore Voting)	294
16	Revise yesterday: Josephus Problem, Block Swap Algorithm (Array Rotation), Leaders in an Array... · Spaced review (3 days ago): Karatsuba's Multiplication, Booth's Algorithm (signed binary multiplication) · Learn: 5.16 Maximum Product Subarray · Practise: 5.16 Maximum Product Subarray · Learn: 5.17 Maximum Equilibrium Sum · Practise: 5.17 Maximum Equilibrium Sum · Learn: 5.18 Maximum Sum of an Hourglass in a Matrix · Practise: 5.18 Maximum Sum of an Hourglass in a Matrix · Learn: 5.19 Toggle the Switch (100 Doors) · Practise: 5.19 Toggle the Switch (100 Doors)	269
17	Revise yesterday: Maximum Product Subarray, Maximum Equilibrium Sum, Maximum Sum of an Hourglass in a Matrix... · Spaced review (3 days ago): Segmented & Incremental Sieve, Binary Palindrome · Learn: 5.20 Alice's Apple Tree (Minimum Garden Perimeter) · Practise: 5.20 Alice's Apple Tree (Minimum Garden Perimeter) · Learn: 5.21 Weighted Substring (Smallest String of a Given Weight) · Practise: 5.21 Weighted Substring (Smallest String of a Given Weight) · Learn: 5.22 Move Hyphen to the Beginning · Practise: 5.22 Move Hyphen to the Beginning · Learn: 5.23 Strobogrammatic Number · Practise: 5.23 Strobogrammatic Number	279
18	Revise yesterday: Alice's Apple Tree (Minimum Garden Perimeter), Weighted Substring (Smallest String of a Given Weight), Move Hyphen to the Beginning... · Spaced review (3 days ago): Josephus Problem, Block Swap Algorithm (Array Rotation) · Learn: 5.24 Lexicographically First Palindromic Rearrangement · Practise: 5.24 Lexicographically First Palindromic Rearrangement · Learn: 5.25 Manacher's Algorithm (Longest Palindromic Substring) · Practise: 5.25 Manacher's Algorithm (Longest Palindromic Substring) · Learn: 5.26 Natural Sort Order · Practise: 5.26 Natural Sort Order · Learn: 5.27 Maneuvering (Counting Grid Paths) · Practise: 5.27 Maneuvering (Counting Grid Paths)	304
19	Revise yesterday: Lexicographically First Palindromic Rearrangement, Manacher's Algorithm (Longest Palindromic Substring), Natural Sort Order... · Spaced review (3 days ago): Maximum Product Subarray, Maximum Equilibrium Sum · Learn: 5.28 Sorted Distinct Permutations · Practise: 5.28 Sorted Distinct Permutations · Learn: 5.29 N-Queens (Backtracking) · Practise: 5.29 N-Queens (Backtracking) · Learn: 5.30 Maze Solving (Rat in a Maze) · Practise: 5.30 Maze Solving (Rat in a Maze)	255
20	Revise yesterday: Sorted Distinct Permutations, N-Queens (Backtracking), Maze Solving (Rat in a Maze) · Spaced review (3 days ago): Alice's Apple Tree (Minimum Garden Perimeter), Weighted Substring (Smallest String of a Given Weight) · Learn: 5.31 Hamiltonian Cycle · Practise: 5.31 Hamiltonian Cycle · Learn: 5.32 Knight's Tour & Warnsdorff's Rule · Practise: 5.32 Knight's Tour & Warnsdorff's Rule · Learn: 5.33 Selection Sort · Practise: 5.33 Selection Sort · Learn: 5.34 Quick Sort · Practise: 5.34 Quick Sort	314
21	Weak-topic review (from your quiz and practice results) · Final mixed assessment (archive topics)	75

30-day plan — minimum-viable path

Day	What to do	Minutes
1	Learn: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Practise: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Learn: 0.3 Installing Java on macOS and Choosing an Editor · Practise: 0.3 Installing Java on macOS and Choosing an Editor · Learn: 0.4 Your First Program: class, main, statements, comments · Practise: 0.4 Your First Program: class, main, statements, comments	130
2	Revise yesterday: How Java Works: Source, Bytecode, JVM, JRE, JDK, Installing Java on macOS and Choosing an Editor, Your First Program: class, main, statements, comments · Learn: 0.5 Compile-time, Runtime and Logic Errors · Practise: 0.5 Compile-time, Runtime and Logic Errors · Learn: 0.6 What Is an Algorithm? · Practise: 0.6 What Is an Algorithm?	110
3	Revise yesterday: Compile-time, Runtime and Logic Errors, What Is an Algorithm? · Learn: 1.1 Variables, Constants and Primitive Types · Practise: 1.1 Variables, Constants and Primitive Types · Learn: 1.2 Type Conversion, Casting and Overflow · Practise: 1.2 Type Conversion, Casting and Overflow	128

Day	What to do	Minutes
4	Revise yesterday: Variables, Constants and Primitive Types, Type Conversion, Casting and Overflow · Spaced review (3 days ago): How Java Works: Source, Bytecode, JVM, JRE, JDK, Installing Java on macOS and Choosing an Editor · Learn: 1.3 Arithmetic, Assignment and Increment Operators · Practise: 1.3 Arithmetic, Assignment and Increment Operators · Learn: 1.4 Relational, Logical and Ternary Operators; Precedence · Practise: 1.4 Relational, Logical and Ternary Operators; Precedence	126
5	Revise yesterday: Arithmetic, Assignment and Increment Operators, Relational, Logical and Ternary Operators; Precedence · Spaced review (3 days ago): Compile-time, Runtime and Logic Errors, What Is an Algorithm? · Learn: 1.5 Bitwise Operators and Shifts · Practise: 1.5 Bitwise Operators and Shifts · Learn: 1.6 Output: print, println, printf and String Concatenation · Practise: 1.6 Output: print, println, printf and String Concatenation	136
6	Revise yesterday: Bitwise Operators and Shifts, Output: print, println, printf and String Concatenation · Spaced review (3 days ago): Variables, Constants and Primitive Types, Type Conversion, Casting and Overflow · Learn: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Practise: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Learn: 1.8 Decision Making: if, else, switch · Practise: 1.8 Decision Making: if, else, switch	136
7	Revise yesterday: Input: Scanner, BufferedReader and Command-Line Arguments, Decision Making: if, else, switch · Spaced review (3 days ago): Arithmetic, Assignment and Increment Operators, Relational, Logical and Ternary Operators; Precedence · Learn: 1.9 Loops: for, while, do-while, break and continue · Practise: 1.9 Loops: for, while, do-while, break and continue · Learn: 1.10 Methods: Parameters, Return Values, Scope and Overloading · Practise: 1.10 Methods: Parameters, Return Values, Scope and Overloading	146
8	Revise yesterday: Loops: for, while, do-while, break and continue, Methods: Parameters, Return Values, Scope and Overloading · Spaced review (3 days ago): Bitwise Operators and Shifts, Output: print, println, printf and String Concatenation · Learn: 1.11 Tracing and Debugging Programs · Practise: 1.11 Tracing and Debugging Programs · Learn: 2.1 One-Dimensional Arrays · Practise: 2.1 One-Dimensional Arrays	136
9	Revise yesterday: Tracing and Debugging Programs, One-Dimensional Arrays · Spaced review (3 days ago): Input: Scanner, BufferedReader and Command-Line Arguments, Decision Making: if, else, switch · Learn: 2.2 Two-Dimensional Arrays (Matrices) · Practise: 2.2 Two-Dimensional Arrays (Matrices) · Learn: 2.3 Strings, StringBuilder and Characters · Practise: 2.3 Strings, StringBuilder and Characters	151
10	Revise yesterday: Two-Dimensional Arrays (Matrices), Strings, StringBuilder and Characters · Spaced review (3 days ago): Loops: for, while, do-while, break and continue, Methods: Parameters, Return Values, Scope and Overloading · Learn: 2.4 Number Systems, Binary and Two's Complement · Practise: 2.4 Number Systems, Binary and Two's Complement · Learn: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic · Practise: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic	151
11	Revise yesterday: Number Systems, Binary and Two's Complement, Divisibility, GCD/LCM, Primes and Modular Arithmetic · Spaced review (3 days ago): Tracing and Debugging Programs, One-Dimensional Arrays · Learn: 2.6 Recursion: Base Case, Recursive Case and the Call Stack · Practise: 2.6 Recursion: Base Case, Recursive Case and the Call Stack · Learn: 2.7 References, Pass-by-Value and Memory · Practise: 2.7 References, Pass-by-Value and Memory	151
12	Revise yesterday: Recursion: Base Case, Recursive Case and the Call Stack, References, Pass-by-Value and Memory · Spaced review (3 days ago): Two-Dimensional Arrays (Matrices), Strings, StringBuilder and Characters · Learn: 2.8 Boundary Conditions, Validation and Common Mistakes · Practise: 2.8 Boundary Conditions, Validation and Common Mistakes · Learn: 3.8 Generics and Lists: ArrayList and LinkedList · Practise: 3.8 Generics and Lists: ArrayList and LinkedList	136
13	Revise yesterday: Boundary Conditions, Validation and Common Mistakes, Generics and Lists: ArrayList and LinkedList · Spaced review (3 days ago): Number Systems, Binary and Two's Complement, Divisibility, GCD/LCM, Primes and Modular Arithmetic · Learn: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Practise: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap	82
14	Revise yesterday: Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Spaced review (3 days ago): Recursion: Base Case, Recursive Case and the Call Stack, References, Pass-by-Value and Memory · Learn: 3.10 Queues, Deques, Stacks and PriorityQueue · Practise: 3.10 Queues, Deques, Stacks and PriorityQueue · Learn: 3.11 Sorting with Comparable and Comparator · Practise: 3.11 Sorting with Comparable and Comparator	136

Day	What to do	Minutes
15	Revise yesterday: Queues, Deques, Stacks and PriorityQueue, Sorting with Comparable and Comparator · Spaced review (3 days ago): Boundary Conditions, Validation and Common Mistakes, Generics and Lists: ArrayList and LinkedList · Learn: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Practise: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Learn: 4.3 Searching: Linear and Binary Search · Practise: 4.3 Searching: Linear and Binary Search	151
16	Revise yesterday: Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables, Searching: Linear and Binary Search · Spaced review (3 days ago): Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Learn: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability · Practise: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability · Learn: 4.5 Merge Sort and Divide and Conquer · Practise: 4.5 Merge Sort and Divide and Conquer	156
17	Revise yesterday: Elementary Sorting: Bubble, Insertion, Selection; Stability, Merge Sort and Divide and Conquer · Spaced review (3 days ago): Queues, Deques, Stacks and PriorityQueue, Sorting with Comparable and Comparator · Learn: 4.7 Time Complexity and Big-O Notation · Practise: 4.7 Time Complexity and Big-O Notation · Learn: 4.8 Algorithm Design Paradigms · Practise: 4.8 Algorithm Design Paradigms	161
18	Revise yesterday: Time Complexity and Big-O Notation, Algorithm Design Paradigms · Spaced review (3 days ago): Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables, Searching: Linear and Binary Search · Learn: 4.9 Analysing Loops: Counting Iterations · Practise: 4.9 Analysing Loops: Counting Iterations	92
19	Revise yesterday: Analysing Loops: Counting Iterations · Spaced review (3 days ago): Elementary Sorting: Bubble, Insertion, Selection; Stability, Merge Sort and Divide and Conquer · Learn: 4.11 Space Complexity and the Call Stack · Practise: 4.11 Space Complexity and the Call Stack · Learn: 5.1 Euclid's Algorithm (GCD) · Practise: 5.1 Euclid's Algorithm (GCD)	136
20	Revise yesterday: Space Complexity and the Call Stack, Euclid's Algorithm (GCD) · Spaced review (3 days ago): Time Complexity and Big-O Notation, Algorithm Design Paradigms · Learn: 5.2 Euler's Phi (Totient) Function · Practise: 5.2 Euler's Phi (Totient) Function · Learn: 5.7 Simple Sieve of Eratosthenes · Practise: 5.7 Simple Sieve of Eratosthenes	151
21	Revise yesterday: Euler's Phi (Totient) Function, Simple Sieve of Eratosthenes · Spaced review (3 days ago): Analysing Loops: Counting Iterations · Learn: 5.9 Binary Palindrome · Practise: 5.9 Binary Palindrome · Learn: 5.12 Josephus Problem · Practise: 5.12 Josephus Problem	151
22	Revise yesterday: Binary Palindrome, Josephus Problem · Spaced review (3 days ago): Space Complexity and the Call Stack, Euclid's Algorithm (GCD) · Learn: 5.13 Block Swap Algorithm (Array Rotation) · Practise: 5.13 Block Swap Algorithm (Array Rotation) · Learn: 5.14 Leaders in an Array · Practise: 5.14 Leaders in an Array	146
23	Revise yesterday: Block Swap Algorithm (Array Rotation), Leaders in an Array · Spaced review (3 days ago): Euler's Phi (Totient) Function, Simple Sieve of Eratosthenes · Learn: 5.15 Majority Element (Boyer–Moore Voting) · Practise: 5.15 Majority Element (Boyer–Moore Voting)	92
24	Revise yesterday: Majority Element (Boyer–Moore Voting) · Spaced review (3 days ago): Binary Palindrome, Josephus Problem · Learn: 5.16 Maximum Product Subarray · Practise: 5.16 Maximum Product Subarray · Learn: 5.17 Maximum Equilibrium Sum · Practise: 5.17 Maximum Equilibrium Sum	151
25	Revise yesterday: Maximum Product Subarray, Maximum Equilibrium Sum · Spaced review (3 days ago): Block Swap Algorithm (Array Rotation), Leaders in an Array · Learn: 5.18 Maximum Sum of an Hourglass in a Matrix · Practise: 5.18 Maximum Sum of an Hourglass in a Matrix · Learn: 5.19 Toggle the Switch (100 Doors) · Practise: 5.19 Toggle the Switch (100 Doors)	136
26	Revise yesterday: Maximum Sum of an Hourglass in a Matrix, Toggle the Switch (100 Doors) · Spaced review (3 days ago): Majority Element (Boyer–Moore Voting) · Learn: 5.22 Move Hyphen to the Beginning · Practise: 5.22 Move Hyphen to the Beginning · Learn: 5.24 Lexicographically First Palindromic Rearrangement · Practise: 5.24 Lexicographically First Palindromic Rearrangement	136

Day	What to do	Minutes
27	Revise yesterday: Move Hyphen to the Beginning, Lexicographically First Palindromic Rearrangement · Spaced review (3 days ago): Maximum Product Subarray, Maximum Equilibrium Sum · Learn: 5.27 Maneuvering (Counting Grid Paths) · Practise: 5.27 Maneuvering (Counting Grid Paths) · Learn: 5.29 N-Queens (Backtracking) · Practise: 5.29 N-Queens (Backtracking)	171
28	Revise yesterday: Maneuvering (Counting Grid Paths), N-Queens (Backtracking) · Spaced review (3 days ago): Maximum Sum of an Hourglass in a Matrix, Toggle the Switch (100 Doors) · Learn: 5.30 Maze Solving (Rat in a Maze) · Practise: 5.30 Maze Solving (Rat in a Maze)	97
29	Revise yesterday: Maze Solving (Rat in a Maze) · Spaced review (3 days ago): Move Hyphen to the Beginning, Lexicographically First Palindromic Rearrangement · Learn: 5.33 Selection Sort · Practise: 5.33 Selection Sort · Learn: 5.34 Quick Sort · Practise: 5.34 Quick Sort	156
30	Weak-topic review (from your quiz and practice results) · Final mixed assessment (archive topics)	75

30-day plan — complete path

Day	What to do	Minutes
1	Learn: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Practise: 0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK · Learn: 0.2 The Java Landscape: History, Features, Editions, JVM Memory · Practise: 0.2 The Java Landscape: History, Features, Editions, JVM Memory · Learn: 0.3 Installing Java on macOS and Choosing an Editor · Practise: 0.3 Installing Java on macOS and Choosing an Editor · Learn: 0.4 Your First Program: class, main, statements, comments · Practise: 0.4 Your First Program: class, main, statements, comments	163
2	Revise yesterday: How Java Works: Source, Bytecode, JVM, JRE, JDK, The Java Landscape: History, Features, Editions, JVM Memory, Installing Java on macOS and Choosing an Editor… · Learn: 0.5 Compile-time, Runtime and Logic Errors · Practise: 0.5 Compile-time, Runtime and Logic Errors · Learn: 0.6 What Is an Algorithm? · Practise: 0.6 What Is an Algorithm? · Learn: 1.1 Variables, Constants and Primitive Types · Practise: 1.1 Variables, Constants and Primitive Types · Learn: 1.2 Type Conversion, Casting and Overflow · Practise: 1.2 Type Conversion, Casting and Overflow	228
3	Revise yesterday: Compile-time, Runtime and Logic Errors, What Is an Algorithm?, Variables, Constants and Primitive Types… · Learn: 1.3 Arithmetic, Assignment and Increment Operators · Practise: 1.3 Arithmetic, Assignment and Increment Operators · Learn: 1.4 Relational, Logical and Ternary Operators; Precedence · Practise: 1.4 Relational, Logical and Ternary Operators; Precedence · Learn: 1.5 Bitwise Operators and Shifts · Practise: 1.5 Bitwise Operators and Shifts	182
4	Revise yesterday: Arithmetic, Assignment and Increment Operators, Relational, Logical and Ternary Operators; Precedence, Bitwise Operators and Shifts · Spaced review (3 days ago): How Java Works: Source, Bytecode, JVM, JRE, JDK, The Java Landscape: History, Features, Editions, JVM Memory · Learn: 1.6 Output: print, println, printf and String Concatenation · Practise: 1.6 Output: print, println, printf and String Concatenation · Learn: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Practise: 1.7 Input: Scanner, BufferedReader and Command-Line Arguments · Learn: 1.8 Decision Making: if, else, switch · Practise: 1.8 Decision Making: if, else, switch	190
5	Revise yesterday: Output: print, println, printf and String Concatenation, Input: Scanner, BufferedReader and Command-Line Arguments, Decision Making: if, else, switch · Spaced review (3 days ago): Compile-time, Runtime and Logic Errors, What Is an Algorithm? · Learn: 1.9 Loops: for, while, do-while, break and continue · Practise: 1.9 Loops: for, while, do-while, break and continue · Learn: 1.10 Methods: Parameters, Return Values, Scope and Overloading · Practise: 1.10 Methods: Parameters, Return Values, Scope and Overloading · Learn: 1.11 Tracing and Debugging Programs · Practise: 1.11 Tracing and Debugging Programs	200
6	Revise yesterday: Loops: for, while, do-while, break and continue, Methods: Parameters, Return Values, Scope and Overloading, Tracing and Debugging Programs · Spaced review (3 days ago): Arithmetic, Assignment and Increment Operators, Relational, Logical and Ternary Operators; Precedence · Learn: 2.1 One-Dimensional Arrays · Practise: 2.1 One-Dimensional Arrays · Learn: 2.2 Two-Dimensional Arrays (Matrices) · Practise: 2.2 Two-Dimensional Arrays (Matrices) · Learn: 2.3 Strings, StringBuilder and Characters · Practise: 2.3 Strings, StringBuilder and Characters	215

Day	What to do	Minutes
7	Revise yesterday: One-Dimensional Arrays, Two-Dimensional Arrays (Matrices), Strings, StringBuilder and Characters · Spaced review (3 days ago): Output: print, println, printf and String Concatenation, Input: Scanner, BufferedReader and Command-Line Arguments · Learn: 2.4 Number Systems, Binary and Two's Complement · Practise: 2.4 Number Systems, Binary and Two's Complement · Learn: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic · Practise: 2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic · Learn: 2.6 Recursion: Base Case, Recursive Case and the Call Stack · Practise: 2.6 Recursion: Base Case, Recursive Case and the Call Stack	225
8	Revise yesterday: Number Systems, Binary and Two's Complement, Divisibility, GCD/LCM, Primes and Modular Arithmetic, Recursion: Base Case, Recursive Case and the Call Stack · Spaced review (3 days ago): Loops: for, while, do-while, break and continue, Methods: Parameters, Return Values, Scope and Overloading · Learn: 2.7 References, Pass-by-Value and Memory · Practise: 2.7 References, Pass-by-Value and Memory · Learn: 2.8 Boundary Conditions, Validation and Common Mistakes · Practise: 2.8 Boundary Conditions, Validation and Common Mistakes · Learn: 3.1 Classes, Objects, Fields, Constructors and this · Practise: 3.1 Classes, Objects, Fields, Constructors and this	195
9	Revise yesterday: References, Pass-by-Value and Memory, Boundary Conditions, Validation and Common Mistakes, Classes, Objects, Fields, Constructors and `this` · Spaced review (3 days ago): One-Dimensional Arrays, Two-Dimensional Arrays (Matrices) · Learn: 3.2 Encapsulation and Access Modifiers · Practise: 3.2 Encapsulation and Access Modifiers · Learn: 3.3 Inheritance, Overriding and Polymorphism · Practise: 3.3 Inheritance, Overriding and Polymorphism	141
10	Revise yesterday: Encapsulation and Access Modifiers, Inheritance, Overriding and Polymorphism · Spaced review (3 days ago): Number Systems, Binary and Two's Complement, Divisibility, GCD/LCM, Primes and Modular Arithmetic · Learn: 3.4 Abstract Classes and Interfaces · Practise: 3.4 Abstract Classes and Interfaces · Learn: 3.5 static, final and Initialisation Order · Practise: 3.5 static, final and Initialisation Order · Learn: 3.6 Packages and Imports · Practise: 3.6 Packages and Imports · Learn: 3.7 Exceptions: try, catch, finally, throw and throws · Practise: 3.7 Exceptions: try, catch, finally, throw and throws	239
11	Revise yesterday: Abstract Classes and Interfaces, static, final and Initialisation Order, Packages and Imports… · Spaced review (3 days ago): References, Pass-by-Value and Memory, Boundary Conditions, Validation and Common Mistakes · Learn: 3.8 Generics and Lists: ArrayList and LinkedList · Practise: 3.8 Generics and Lists: ArrayList and LinkedList · Learn: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Practise: 3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Learn: 3.10 Queues, Deques, Stacks and PriorityQueue · Practise: 3.10 Queues, Deques, Stacks and PriorityQueue	205
12	Revise yesterday: Generics and Lists: ArrayList and LinkedList, Sets and Maps: HashSet, TreeSet, HashMap, TreeMap, Queues, Deques, Stacks and PriorityQueue · Spaced review (3 days ago): Encapsulation and Access Modifiers, Inheritance, Overriding and Polymorphism · Learn: 3.11 Sorting with Comparable and Comparator · Practise: 3.11 Sorting with Comparable and Comparator · Learn: 3.12 equals(), hashCode() and Immutability · Practise: 3.12 equals(), hashCode() and Immutability · Learn: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Practise: 4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables	205
13	Revise yesterday: Sorting with Comparable and Comparator, equals(), hashCode() and Immutability, Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables · Spaced review (3 days ago): Abstract Classes and Interfaces, static, final and Initialisation Order · Learn: 4.2 Trees, Graphs and Heaps · Practise: 4.2 Trees, Graphs and Heaps · Learn: 4.3 Searching: Linear and Binary Search · Practise: 4.3 Searching: Linear and Binary Search	156
14	Revise yesterday: Trees, Graphs and Heaps, Searching: Linear and Binary Search · Spaced review (3 days ago): Generics and Lists: ArrayList and LinkedList, Sets and Maps: HashSet, TreeSet, HashMap, TreeMap · Learn: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability · Practise: 4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability · Learn: 4.5 Merge Sort and Divide and Conquer · Practise: 4.5 Merge Sort and Divide and Conquer · Learn: 4.6 Iteration versus Recursion · Practise: 4.6 Iteration versus Recursion	215

Day	What to do	Minutes
15	Revise yesterday: Elementary Sorting: Bubble, Insertion, Selection; Stability, Merge Sort and Divide and Conquer, Iteration versus Recursion · Spaced review (3 days ago): Sorting with Comparable and Comparator, equals(), hashCode() and Immutability · Learn: 4.7 Time Complexity and Big-O Notation · Practise: 4.7 Time Complexity and Big-O Notation · Learn: 4.8 Algorithm Design Paradigms · Practise: 4.8 Algorithm Design Paradigms · Learn: 4.9 Analysing Loops: Counting Iterations · Practise: 4.9 Analysing Loops: Counting Iterations	235
16	Revise yesterday: Time Complexity and Big-O Notation, Algorithm Design Paradigms, Analysing Loops: Counting Iterations · Spaced review (3 days ago): Trees, Graphs and Heaps, Searching: Linear and Binary Search · Learn: 4.10 Analysing Recursion: Recurrence Relations · Practise: 4.10 Analysing Recursion: Recurrence Relations · Learn: 4.11 Space Complexity and the Call Stack · Practise: 4.11 Space Complexity and the Call Stack · Learn: 4.12 Best, Average and Worst Case · Practise: 4.12 Best, Average and Worst Case	190
17	Revise yesterday: Analysing Recursion: Recurrence Relations, Space Complexity and the Call Stack, Best, Average and Worst Case · Spaced review (3 days ago): Elementary Sorting: Bubble, Insertion, Selection; Stability, Merge Sort and Divide and Conquer · Learn: 5.1 Euclid's Algorithm (GCD) · Practise: 5.1 Euclid's Algorithm (GCD) · Learn: 5.2 Euler's Phi (Totient) Function · Practise: 5.2 Euler's Phi (Totient) Function · Learn: 5.3 Chinese Remainder Theorem · Practise: 5.3 Chinese Remainder Theorem	220
18	Revise yesterday: Euclid's Algorithm (GCD), Euler's Phi (Totient) Function, Chinese Remainder Theorem · Spaced review (3 days ago): Time Complexity and Big-O Notation, Algorithm Design Paradigms · Learn: 5.4 Combinations (nCr) · Practise: 5.4 Combinations (nCr) · Learn: 5.5 Karatsuba's Multiplication · Practise: 5.5 Karatsuba's Multiplication	161
19	Revise yesterday: Combinations (nCr), Karatsuba's Multiplication · Spaced review (3 days ago): Analysing Recursion: Recurrence Relations, Space Complexity and the Call Stack · Learn: 5.6 Booth's Algorithm (signed binary multiplication) · Practise: 5.6 Booth's Algorithm (signed binary multiplication) · Learn: 5.7 Simple Sieve of Eratosthenes · Practise: 5.7 Simple Sieve of Eratosthenes · Learn: 5.8 Segmented & Incremental Sieve · Practise: 5.8 Segmented & Incremental Sieve	245
20	Revise yesterday: Booth's Algorithm (signed binary multiplication), Simple Sieve of Eratosthenes, Segmented & Incremental Sieve · Spaced review (3 days ago): Euclid's Algorithm (GCD), Euler's Phi (Totient) Function · Learn: 5.9 Binary Palindrome · Practise: 5.9 Binary Palindrome · Learn: 5.10 Swap Two Nibbles in a Byte · Practise: 5.10 Swap Two Nibbles in a Byte	131
21	Revise yesterday: Binary Palindrome, Swap Two Nibbles in a Byte · Spaced review (3 days ago): Combinations (nCr), Karatsuba's Multiplication · Learn: 5.11 Longest Sequence of 1s (and after flipping a bit) · Practise: 5.11 Longest Sequence of 1s (and after flipping a bit) · Learn: 5.12 Josephus Problem · Practise: 5.12 Josephus Problem · Learn: 5.13 Block Swap Algorithm (Array Rotation) · Practise: 5.13 Block Swap Algorithm (Array Rotation)	235
22	Revise yesterday: Longest Sequence of 1s (and after flipping a bit), Josephus Problem, Block Swap Algorithm (Array Rotation) · Spaced review (3 days ago): Booth's Algorithm (signed binary multiplication), Simple Sieve of Eratosthenes · Learn: 5.14 Leaders in an Array · Practise: 5.14 Leaders in an Array · Learn: 5.15 Majority Element (Boyer–Moore Voting) · Practise: 5.15 Majority Element (Boyer–Moore Voting) · Learn: 5.16 Maximum Product Subarray · Practise: 5.16 Maximum Product Subarray	220
23	Revise yesterday: Leaders in an Array, Majority Element (Boyer–Moore Voting), Maximum Product Subarray · Spaced review (3 days ago): Binary Palindrome, Swap Two Nibbles in a Byte · Learn: 5.17 Maximum Equilibrium Sum · Practise: 5.17 Maximum Equilibrium Sum · Learn: 5.18 Maximum Sum of an Hourglass in a Matrix · Practise: 5.18 Maximum Sum of an Hourglass in a Matrix · Learn: 5.19 Toggle the Switch (100 Doors) · Practise: 5.19 Toggle the Switch (100 Doors)	195
24	Revise yesterday: Maximum Equilibrium Sum, Maximum Sum of an Hourglass in a Matrix, Toggle the Switch (100 Doors) · Spaced review (3 days ago): Longest Sequence of 1s (and after flipping a bit), Josephus Problem · Learn: 5.20 Alice's Apple Tree (Minimum Garden Perimeter) · Practise: 5.20 Alice's Apple Tree (Minimum Garden Perimeter) · Learn: 5.21 Weighted Substring (Smallest String of a Given Weight) · Practise: 5.21 Weighted Substring (Smallest String of a Given Weight)	161

Day	What to do	Minutes
25	Revise yesterday: Alice's Apple Tree (Minimum Garden Perimeter), Weighted Substring (Smallest String of a Given Weight) · Spaced review (3 days ago): Leaders in an Array, Majority Element (Boyer–Moore Voting) · Learn: 5.22 Move Hyphen to the Beginning · Practise: 5.22 Move Hyphen to the Beginning · Learn: 5.23 Strobogrammatic Number · Practise: 5.23 Strobogrammatic Number · Learn: 5.24 Lexicographically First Palindromic Rearrangement · Practise: 5.24 Lexicographically First Palindromic Rearrangement	200
26	Revise yesterday: Move Hyphen to the Beginning, Strobogrammatic Number, Lexicographically First Palindromic Rearrangement · Spaced review (3 days ago): Maximum Equilibrium Sum, Maximum Sum of an Hourglass in a Matrix · Learn: 5.25 Manacher's Algorithm (Longest Palindromic Substring) · Practise: 5.25 Manacher's Algorithm (Longest Palindromic Substring) · Learn: 5.26 Natural Sort Order · Practise: 5.26 Natural Sort Order · Learn: 5.27 Maneuvering (Counting Grid Paths) · Practise: 5.27 Maneuvering (Counting Grid Paths)	240
27	Revise yesterday: Manacher's Algorithm (Longest Palindromic Substring), Natural Sort Order, Maneuvering (Counting Grid Paths) · Spaced review (3 days ago): Alice's Apple Tree (Minimum Garden Perimeter), Weighted Substring (Smallest String of a Given Weight) · Learn: 5.28 Sorted Distinct Permutations · Practise: 5.28 Sorted Distinct Permutations · Learn: 5.29 N-Queens (Backtracking) · Practise: 5.29 N-Queens (Backtracking)	176
28	Revise yesterday: Sorted Distinct Permutations, N-Queens (Backtracking) · Spaced review (3 days ago): Move Hyphen to the Beginning, Strobogrammatic Number · Learn: 5.30 Maze Solving (Rat in a Maze) · Practise: 5.30 Maze Solving (Rat in a Maze) · Learn: 5.31 Hamiltonian Cycle · Practise: 5.31 Hamiltonian Cycle	176
29	Revise yesterday: Maze Solving (Rat in a Maze), Hamiltonian Cycle · Spaced review (3 days ago): Manacher's Algorithm (Longest Palindromic Substring), Natural Sort Order · Learn: 5.32 Knight's Tour & Warnsdorff's Rule · Practise: 5.32 Knight's Tour & Warnsdorff's Rule · Learn: 5.33 Selection Sort · Practise: 5.33 Selection Sort · Learn: 5.34 Quick Sort · Practise: 5.34 Quick Sort	235
30	Weak-topic review (from your quiz and practice results) · Final mixed assessment (archive topics)	75

AN HONEST PROMISE

This guide gives you a complete, verified map and enough practice to learn the material quickly — but nobody can promise mastery in a fixed number of days. How fast you progress depends on how much you code yourself. Use the quizzes and exercises to find out what you actually know, and revisit weak topics.

Progress checklist

0 · Programming & Java Setup

	Lesson	Core	Min	Mastery check
☐	0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK	★	25	☐ quiz ≥ 80% ☐ exercises done
☐	0.2 The Java Landscape: History, Features, Editions, JVM Memory		25	☐ quiz ≥ 80% ☐ exercises done
☐	0.3 Installing Java on macOS and Choosing an Editor	★	30	☐ quiz ≥ 80% ☐ exercises done
☐	0.4 Your First Program: class, main, statements, comments	★	35	☐ quiz ≥ 80% ☐ exercises done
☐	0.5 Compile-time, Runtime and Logic Errors	★	30	☐ quiz ≥ 80% ☐ exercises done
☐	0.6 What Is an Algorithm?	★	30	☐ quiz ≥ 80% ☐ exercises done

1 · Java Syntax & Fundamentals

	Lesson	Core	Min	Mastery check
☐	1.1 Variables, Constants and Primitive Types	★	35	☐ quiz ≥ 80% ☐ exercises done
☐	1.2 Type Conversion, Casting and Overflow	★	35	☐ quiz ≥ 80% ☐ exercises done
☐	1.3 Arithmetic, Assignment and Increment Operators	★	30	☐ quiz ≥ 80% ☐ exercises done
☐	1.4 Relational, Logical and Ternary Operators; Precedence	★	30	☐ quiz ≥ 80% ☐ exercises done
☐	1.5 Bitwise Operators and Shifts	★	40	☐ quiz ≥ 80% ☐ exercises done
☐	1.6 Output: print, println, printf and String Concatenation	★	30	☐ quiz ≥ 80% ☐ exercises done
☐	1.7 Input: Scanner, BufferedReader and Command-Line Arguments	★	35	☐ quiz ≥ 80% ☐ exercises done
☐	1.8 Decision Making: if, else, switch	★	35	☐ quiz ≥ 80% ☐ exercises done
☐	1.9 Loops: for, while, do-while, break and continue	★	40	☐ quiz ≥ 80% ☐ exercises done
☐	1.10 Methods: Parameters, Return Values, Scope and Overloading	★	40	☐ quiz ≥ 80% ☐ exercises done
☐	1.11 Tracing and Debugging Programs	★	30	☐ quiz ≥ 80% ☐ exercises done

2 · Core Programming Skills

	Lesson	Core	Min	Mastery check
☐	2.1 One-Dimensional Arrays	★	40	☐ quiz ≥ 80% ☐ exercises done
☐	2.2 Two-Dimensional Arrays (Matrices)	★	40	☐ quiz ≥ 80% ☐ exercises done
☐	2.3 Strings, StringBuilder and Characters	★	45	☐ quiz ≥ 80% ☐ exercises done
☐	2.4 Number Systems, Binary and Two's Complement	★	40	☐ quiz ≥ 80% ☐ exercises done
☐	2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic	★	45	☐ quiz ≥ 80% ☐ exercises done
☐	2.6 Recursion: Base Case, Recursive Case and the Call Stack	★	50	☐ quiz ≥ 80% ☐ exercises done

	Lesson	Core	Min	Mastery check
☐	2.7 References, Pass-by-Value and Memory	★	35	☐ quiz ≥ 80% ☐ exercises done
☐	2.8 Boundary Conditions, Validation and Common Mistakes	★	30	☐ quiz ≥ 80% ☐ exercises done

3 · Object-Oriented Programming

	Lesson	Core	Min	Mastery check
☐	3.1 Classes, Objects, Fields, Constructors and this		40	☐ quiz ≥ 80% ☐ exercises done
☐	3.2 Encapsulation and Access Modifiers		30	☐ quiz ≥ 80% ☐ exercises done
☐	3.3 Inheritance, Overriding and Polymorphism		45	☐ quiz ≥ 80% ☐ exercises done
☐	3.4 Abstract Classes and Interfaces		40	☐ quiz ≥ 80% ☐ exercises done
☐	3.5 static, final and Initialisation Order		30	☐ quiz ≥ 80% ☐ exercises done
☐	3.6 Packages and Imports		20	☐ quiz ≥ 80% ☐ exercises done
☐	3.7 Exceptions: try, catch, finally, throw and throws		35	☐ quiz ≥ 80% ☐ exercises done
☐	3.8 Generics and Lists: ArrayList and LinkedList	★	40	☐ quiz ≥ 80% ☐ exercises done
☐	3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap	★	40	☐ quiz ≥ 80% ☐ exercises done
☐	3.10 Queues, Deques, Stacks and PriorityQueue	★	35	☐ quiz ≥ 80% ☐ exercises done
☐	3.11 Sorting with Comparable and Comparator	★	35	☐ quiz ≥ 80% ☐ exercises done
☐	3.12 equals(), hashCode() and Immutability		35	☐ quiz ≥ 80% ☐ exercises done

4 · Data Structures & Algorithm Foundations

	Lesson	Core	Min	Mastery check
☐	4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables	★	45	☐ quiz ≥ 80% ☐ exercises done
☐	4.2 Trees, Graphs and Heaps		50	☐ quiz ≥ 80% ☐ exercises done
☐	4.3 Searching: Linear and Binary Search	★	40	☐ quiz ≥ 80% ☐ exercises done
☐	4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability	★	45	☐ quiz ≥ 80% ☐ exercises done
☐	4.5 Merge Sort and Divide and Conquer	★	45	☐ quiz ≥ 80% ☐ exercises done
☐	4.6 Iteration versus Recursion		35	☐ quiz ≥ 80% ☐ exercises done
☐	4.7 Time Complexity and Big-O Notation	★	45	☐ quiz ≥ 80% ☐ exercises done
☐	4.8 Algorithm Design Paradigms	★	50	☐ quiz ≥ 80% ☐ exercises done
☐	4.9 Analysing Loops: Counting Iterations	★	50	☐ quiz ≥ 80% ☐ exercises done
☐	4.10 Analysing Recursion: Recurrence Relations		40	☐ quiz ≥ 80% ☐ exercises done
☐	4.11 Space Complexity and the Call Stack	★	30	☐ quiz ≥ 80% ☐ exercises done
☐	4.12 Best, Average and Worst Case		30	☐ quiz ≥ 80% ☐ exercises done

5 · Archive Algorithm Mastery

	Lesson	Core	Min	Mastery check
☐	5.1 Euclid's Algorithm (GCD)	★	40	☐ quiz ≥ 80% ☐ exercises done
☐	5.2 Euler's Phi (Totient) Function	★	45	☐ quiz ≥ 80% ☐ exercises done
☐	5.3 Chinese Remainder Theorem		45	☐ quiz ≥ 80% ☐ exercises done
☐	5.4 Combinations (nCr)		40	☐ quiz ≥ 80% ☐ exercises done
☐	5.5 Karatsuba's Multiplication		55	☐ quiz ≥ 80% ☐ exercises done
☐	5.6 Booth's Algorithm (signed binary multiplication)		60	☐ quiz ≥ 80% ☐ exercises done
☐	5.7 Simple Sieve of Eratosthenes	★	40	☐ quiz ≥ 80% ☐ exercises done
☐	5.8 Segmented & Incremental Sieve		55	☐ quiz ≥ 80% ☐ exercises done
☐	5.9 Binary Palindrome	★	35	☐ quiz ≥ 80% ☐ exercises done
☐	5.10 Swap Two Nibbles in a Byte		30	☐ quiz ≥ 80% ☐ exercises done
☐	5.11 Longest Sequence of 1s (and after flipping a bit)		45	☐ quiz ≥ 80% ☐ exercises done
☐	5.12 Josephus Problem	★	50	☐ quiz ≥ 80% ☐ exercises done
☐	5.13 Block Swap Algorithm (Array Rotation)	★	50	☐ quiz ≥ 80% ☐ exercises done
☐	5.14 Leaders in an Array	★	30	☐ quiz ≥ 80% ☐ exercises done
☐	5.15 Majority Element (Boyer–Moore Voting)	★	50	☐ quiz ≥ 80% ☐ exercises done
☐	5.16 Maximum Product Subarray	★	50	☐ quiz ≥ 80% ☐ exercises done
☐	5.17 Maximum Equilibrium Sum	★	35	☐ quiz ≥ 80% ☐ exercises done
☐	5.18 Maximum Sum of an Hourglass in a Matrix	★	30	☐ quiz ≥ 80% ☐ exercises done
☐	5.19 Toggle the Switch (100 Doors)	★	40	☐ quiz ≥ 80% ☐ exercises done
☐	5.20 Alice's Apple Tree (Minimum Garden Perimeter)		45	☐ quiz ≥ 80% ☐ exercises done
☐	5.21 Weighted Substring (Smallest String of a Given Weight)		50	☐ quiz ≥ 80% ☐ exercises done
☐	5.22 Move Hyphen to the Beginning	★	30	☐ quiz ≥ 80% ☐ exercises done
☐	5.23 Strobogrammatic Number		40	☐ quiz ≥ 80% ☐ exercises done
☐	5.24 Lexicographically First Palindromic Rearrangement	★	40	☐ quiz ≥ 80% ☐ exercises done
☐	5.25 Manacher's Algorithm (Longest Palindromic Substring)		65	☐ quiz ≥ 80% ☐ exercises done
☐	5.26 Natural Sort Order		40	☐ quiz ≥ 80% ☐ exercises done
☐	5.27 Maneuvering (Counting Grid Paths)	★	45	☐ quiz ≥ 80% ☐ exercises done
☐	5.28 Sorted Distinct Permutations		50	☐ quiz ≥ 80% ☐ exercises done
☐	5.29 N-Queens (Backtracking)	★	60	☐ quiz ≥ 80% ☐ exercises done
☐	5.30 Maze Solving (Rat in a Maze)	★	55	☐ quiz ≥ 80% ☐ exercises done
☐	5.31 Hamiltonian Cycle		55	☐ quiz ≥ 80% ☐ exercises done
☐	5.32 Knight's Tour & Warnsdorff's Rule		55	☐ quiz ≥ 80% ☐ exercises done

	Lesson	Core	Min	Mastery check
☐	5.33 Selection Sort	★	35	☐ quiz ≥ 80% ☐ exercises done
☐	5.34 Quick Sort	★	55	☐ quiz ≥ 80% ☐ exercises done